You

I have my spirits exam for tech associate role in de shaw. I'll give you the topics and youll generate a cheat sheet for all the formulas, how to solve each variations and every detail i need to know.

Gemini 3.6 Thinking

D. E. Shaw is known for setting a high bar—their tech assessments heavily test multi-step quantitative reasoning, probability/combinatorics, dynamic programming, and GATE-level CS core fundamentals (OS, DBMS, Networks, DSA).

Once you share your list of topics, I will generate a comprehensive cheat sheet engineered for maximum clarity and quick revision.

What I'll Build for Each Topic

- **Core Formulas & Mathematical Proofs:** Equations (rendered in clear LaTeX) and fundamental theorems.
- **Pattern Recognition & Problem Variations:** How to identify the exact variation of a problem in under 15 seconds.
- **Step-by-Step Solving Blueprints:** Systematic methods to solve each pattern rapidly under timed pressure.
- **Edge Cases & Common Pitfalls:** Tricky constraints, boundary conditions, and negative-marking traps to watch out for.

Drop your topic list whenever you're ready—are there specific sections (e.g., Quant/Probability, Dynamic Programming, or CS Fundamentals) you want to prioritize first?

You

Percentages and profit and loss: percentage-change scenarios layered with profit and loss

Gemini 3.6 Thinking

D. E. Shaw Quantitative Cheat Sheet

Percentages & Layered Profit & Loss

In high-speed quantitative tests like D. E. Shaw's, traditional algebraic equations (x and y) will slow you down. The key to cracking layered percentage and profit/loss problems is **Multiplier Algebra (K-Factors)** and **Base-100 Shifts**.

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1. Core Multiplier Formulas & Master Equations

The Multiplier Method (K-Factor)

Convert all percentage changes into direct decimal multipliers:

- **Increase of $x\%$:** $K = 1 + \frac{x}{100}$
- **Decrease / Discount of $x\%$:** $K = 1 - \frac{x}{100}$
- **Net Selling Price:** $SP = CP \times K_{profit} = MP \times K_{discount}$

Successive Percentage Change Formula

For two successive percentage changes of $a\%$ and $b\%$:

$$\text{Net } \% \text{ Change} = a + b + \frac{a \times b}{100}$$

> **Note:** Assign a positive sign $(+)$ for markups/profits and a negative sign $(-)$ for discounts/losses.

The Master Markup-Discount-Profit Relation

Eliminate the need to calculate intermediate prices using this direct ratio:

$$\frac{MP}{CP} = \frac{100 + P\%}{100 - D\%}$$

If the transaction results in a loss of $L\%$:

$$\frac{MP}{CP} = \frac{100 - L\%}{100 - D\%}$$

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2. Problem Variations & Rapid Blueprints

Variation 1: Multi-Stage Layering (Markup $\rightarrow$ Discount $\rightarrow$ Tax / Profit)

Scenario: An item is marked up by $M\%$, given successive discounts of $D_1\%$ and $D_2\%$, and subjected to a tax/fee of $T\%$.

Blueprint: Chain Multiplier

1. Write the net multiplier equation directly:

$$SP = CP \times \left(1 + \frac{M}{100}\right) \times \left(1 - \frac{D_1}{100}\right) \times \left(1 - \frac{D_2}{100}\right) \times \left(1 + \frac{T}{100}\right)$$

2. Profit % is given by $(K_{\text{net}} - 1) \times 100\%$.

Variation 2: Dynamic CP & SP Shifts (Hypothetical Re-runs)

Scenario: *"A trader sells an article at $P_1\%$ profit. Had he bought it for $x\%$ less and sold it for \R $less/more, he would have made a profit of P_2\%$. Find the original CP."*

Blueprint: Scale Variable Method

1. Set initial $CP = 100u$.
2. Write initial $SP = (100 + P_1)u$.
3. Compute hypothetical $CP' = 100u \times \left(1 - \frac{x}{100}\right)$.
4. Compute hypothetical SP' using $P_2\%$:

$$SP' = CP' \times \left(1 + \frac{P_2}{100}\right)$$

5. Equate the difference between SP and SP' to the absolute dollar amount R:

$$|SP - SP'| = R$$

6. Solve for u. Original $CP = 100u$.

Variation 3: Dishonest Trader Layered with Percentage Changes

Scenario: A dealer marks up goods by $M\%$, offers a discount of $D\%$, but uses a false weight (gives W_{actual} instead of claiming W_{claimed}).

Blueprint: Compound Ratio Multiplier

Separate price manipulation from weight fraud into distinct multipliers:

$$\text{Price Multiplier } (K_{\text{price}}) = \left(1 + \frac{M}{100}\right) \times \left(1 - \frac{D}{100}\right)$$

$$\text{Weight Multiplier } (K_{\text{weight}}) = \frac{\text{Claimed Weight}}{\text{Actual Weight Given}} = \frac{W_{\text{claimed}}}{W_{\text{actual}}}$$

$$\text{Net Multiplier } (K_{\text{net}}) = K_{\text{price}} \times K_{\text{weight}}$$

$$\text{Net Profit } \% = (K_{\text{net}} - 1) \times 100$$

> **Rule of Thumb:** Fraudulent weight *always* increases the net multiplier. If the dealer gives 800g instead of 1000g, $K_{\text{weight}} = \frac{1000}{800} = 1.25$.

Variation 4: Inventory Split (Weighted Average Profit/Loss)

Scenario: A fraction f_1 of inventory is sold at $P_1\%$ profit, fraction f_2 at $P_2\%$ profit, and the remainder f_3 at a loss of $L\%$.

Blueprint: Direct Linear Combination

$$P_{\text{overall}}\% = (f_1 \times P_1) + (f_2 \times P_2) - (f_3 \times L)$$

- Ensure $f_1 + f_2 + f_3 = 1$.
- If $P_{\text{overall}}\%$ is negative, the net transaction is a loss.

Variation 5: Dual Item Transactions (Equal SP or Equal CP)

Condition	Given Constraints	Net Result Formula	Notes
Same Selling Price ($SP_1 = SP_2$**)**	Item 1: $+x\%$ profit Item 2: $-x\%$ loss	**Always Loss:** $\text{Loss } \% = \left(\frac{x}{10}\right)^2$	Never breaks even.
Same Selling Price ($SP_1 = SP_2$**)**	Item 1: $+a\%$ profit Item 2: $-b\%$ loss	$\text{Net } \% = \left(\frac{100(a-b)-2ab}{200+a-b}\right)$	Use formulas for exact values; avoid approximations.
Same Cost Price ($CP_1 = CP_2$**)**	Item 1: $+x\%$ profit Item 2: $-x\%$ loss	**Net Result = 0%** (Break Even)	Profits cancel out exactly.

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3. Fast Mental Calculation Shortcuts

Fraction-to-Percentage Equivalent Reference

Fraction	Percentage	Fraction	Percentage
$\frac{1}{6}$	16.67%	$\frac{1}{9}$	11.11%
$\frac{1}{7}$	14.28%	$\frac{1}{11}$	9.09%
$\frac{1}{8}$	12.50%	$\frac{1}{12}$	8.33%

The "Base Shift" Principle

- If A is $x\%$ more than B, then B is **less** than A by:

$$\left(\frac{x}{100+x}\right) \times 100\%$$

- If A is $x\%$ less than B, then B is **more** than A by:

$$\left(\frac{x}{100-x}\right) \times 100\%$$

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4. Common Traps & Edge Cases

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[ TRAP ALERT 1: Base Misplacement ]
• Markup % is ALWAYS calculated on CP.
• Discount % is ALWAYS calculated on MP.
• Profit/Loss % is calculated on CP (unless explicitly specified as "Profit on SP").
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- **Profit on Selling Price Trap:** If a question specifies *"Profit of $x\%$ on Selling Price"*:

$$\text{Actual Profit \% (on CP)} = \left(\frac{x}{100-x}\right) \times 100$$

- **False Weight Base Trap:** When calculating profit percentage from false weights alone, the denominator is always the **actual weight delivered to the customer**, not the claimed weight:

$$\text{Profit \%} = \left(\frac{\text{Error Weight}}{\text{Actual Weight Delivered}}\right) \times 100$$

- **Percentage Points vs. Percentage Change:** If profit increases from 20% to 25%, that is an increase of 5 **percentage points**, but a 25% **percentage increase** in profit margin ($\frac{25-20}{20} \times 100$). Read questions carefully to avoid mixing these up.

You

Time, speed, and distance: boats and streams, trains, and multi-speed problems

Gemini 3.6 Thinking

D. E. Shaw Quantitative Cheat Sheet

Time, Speed, & Distance: Boats, Trains, & Multi-Speed Mechanics

To tackle Time, Speed, and Distance (TSD) questions at D. E. Shaw speed, avoid solving multi-variable systems from scratch. Utilize **Ratio Proportionality**, **Relative Speed Vectors**, and **Harmonic/Arithmetic Means**.

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1. Core Relations & Unit Conversions

Fundamental Proportionality Rules

When one variable remains constant, use direct ratios to solve in seconds:

- **Constant Distance (d is fixed):** Speed and Time are inversely proportional.

$$\frac{v_1}{v_2} = \frac{t_2}{t_1}$$

- **Constant Time (t is fixed):** Distance and Speed are directly proportional.

$$\frac{d_1}{d_2} = \frac{v_1}{v_2}$$

- **Constant Speed (v is fixed):** Distance and Time are directly proportional.

$$\frac{d_1}{d_2} = \frac{t_1}{t_2}$$

Lightning Unit Conversions

- $\mathrm{km/h}$ **to** $\mathrm{m/s}$**:** Multiply by $\frac{5}{18}$
- $\mathrm{m/s}$ **to** $\mathrm{km/h}$**:** Multiply by $\frac{18}{5}$
- **Quick Benchmark Values:**
  - $18\ \mathrm{km/h} = 5\ \mathrm{m/s}$
  - $36\ \mathrm{km/h} = 10\ \mathrm{m/s}$
  - $54\ \mathrm{km/h} = 15\ \mathrm{m/s}$
  - $72\ \mathrm{km/h} = 20\ \mathrm{m/s}$
  - $90\ \mathrm{km/h} = 25\ \mathrm{m/s}$

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2. Boats and Streams

Let:

- $u =$ Speed of boat in still water
- $v =$ Speed of the stream/current
- $S_d =$ Downstream speed (with current) $= u + v$
- $S_u =$ Upstream speed (against current) $= u - v$

Core Derived Equations

$$u = \frac{S_d + S_u}{2}$$

$$v = \frac{S_d - S_u}{2}$$

High-Frequency Variations

Variation 1: Round Trip Equal Distance

Scenario: A boat goes to a destination at distance d and returns to the starting point.

- **Total Time Taken (T):**

$$T = \frac{d}{u+v} + \frac{d}{u-v} = \frac{2du}{u^2 - v^2}$$

- **Distance (d):**

$$d = \frac{T(u^2 - v^2)}{2u}$$

Variation 2: Ratio of Upstream to Downstream Time

Scenario: A boat takes k times as long to row upstream as to row downstream for the same distance ($t_u = k \cdot t_d$).

Using Componendo and Dividendo on $\frac{u+v}{u-v} = \frac{k}{1}$:

$\frac{u}{v} = \frac{k+1}{k-1} = \frac{t_u+t_d}{t_u-t_d}$

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3. Train Mechanics & Relative Speed

Basic Crossing Equations

$\text{Time } (t) = \frac{\text{Total Distance to Cover}}{\text{Relative Speed}}$

Scenario	Total Distance	Relative Speed (S_{rel})	Formula
Train passing stationary point (Pole, Signal, Standing Man)	L_{train}	v_{train}	$t = \frac{L_{\text{train}}}{v_{\text{train}}}$
Train passing stationary object with length (Platform, Bridge, Tunnel)	$L_{\text{train}} + L_{\text{object}}$	v_{train}	$t = \frac{L_{\text{train}}+L_{\text{object}}}{v_{\text{train}}}$
Two trains in OPPOSITE directions	$L_1 + L_2$	$v_1 + v_2$	$t = \frac{L_1+L_2}{v_1+v_2}$
Two trains in SAME direction	$L_1 + L_2$	$\|v_1 - v_2\|$	$t = \frac{L_1+L_2}{\|v_1-v_2\|}$

Tricky Train Variations

Variation 1: Train Passing a Person Inside Another Train

Scenario: Train A (length L_1, speed v_1) passes a passenger sitting inside Train B (length L_2, speed v_2).

- **Key Insight:** Ignore L_2. The distance covered is strictly L_1 (the passing train's length).

$t = \frac{L_1}{v_1 \pm v_2}$

Variation 2: Staggered Departure Times (Chasing / Meeting)

Scenario: Train 1 leaves Station A at T_1 with speed v_1. Train 2 leaves Station A at T_2 ($T_2 > T_1$) with speed v_2 ($v_2 > v_1$).

1. Calculate initial head start distance of Train 1:

   $d_{\text{lead}} = v_1 \times (T_2 - T_1)$

2. Time taken for Train 2 to catch Train 1 from time T_2:

   $t_{\text{catch}} = \frac{d_{\text{lead}}}{v_2 - v_1}$

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4. Multi-Speed & Dynamic Scenarios

Average Speed Equations

$\text{Average Speed } (S_{\text{avg}}) = \frac{\text{Total Distance}}{\text{Total Time}}$

- **Case 1: Equal Distances Covered** ($d_1 = d_2 = d$) (Harmonic Mean)

$S_{\text{avg}} = \frac{2v_1v_2}{v_1+v_2}$

- **Case 2: Three Equal Distances** ($d_1 = d_2 = d_3$)

$S_{\text{avg}} = \frac{3v_1v_2v_3}{v_1v_2+v_2v_3+v_3v_1}$

- **Case 3: Equal Time Intervals** ($t_1 = t_2 = t$) (Arithmetic Mean)

$S_{\text{avg}} = \frac{v_1 + v_2}{2}$

The Early / Late Shortcut (Delta-Time Formula)

Scenario: If a person travels at speed v_1, they are t_1 minutes late. If they travel at speed v_2, they are t_2 minutes early.

$$\text{Distance } (d) = \frac{v_1 \cdot v_2}{|v_1 - v_2|} \times \Delta t$$

> **Rule for Δt:**
>
> - Late + Early $\rightarrow$ Add times ($\Delta t = t_1 + t_2$)
> - Late + Late OR Early + Early $\rightarrow$ Subtract times ($\Delta t = |t_1 - t_2|$)
> - *Ensure Δt is converted to hours if speeds are in km/h.*

Stoppage Time / Delays

Scenario: A train travels at speed V_{excl} without stoppages and V_{incl} with stoppages.

$$\text{Stoppage Time per Hour} = \frac{V_{\text{excl}} - V_{\text{incl}}}{V_{\text{excl}}} \times 60 \text{ minutes}$$

Post-Meeting Speed Ratio Formula (Inverse Square Root Law)

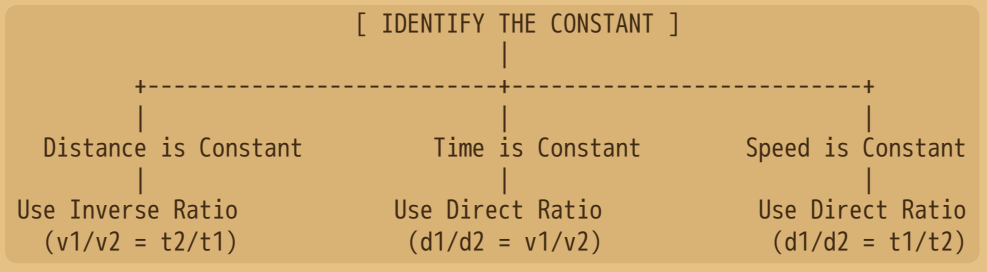
Scenario: Two bodies start simultaneously from points A and B towards each other. After meeting on the way, they take t_A and t_B time to reach B and A respectively.

$$\frac{v_A}{v_B} = \sqrt{\frac{t_B}{t_A}}$$

$$\text{Time taken to meet initially } (t_{\text{meet}}) = \sqrt{t_A \cdot t_B}$$

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5. High-Speed Decision Tree & Traps




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[ TRAP ALERT 1: Unit Discrepancies ]
• Watch out for speeds in km/h mixed with train lengths in meters and times in seconds.
• Always convert speeds to m/s early: multiply by (5/18).

[ TRAP ALERT 2: River Current Direction ]
• Read carefully whether the boat starts upstream or downstream.
• "Moving with the current" = Downstream (u + v).
• "Moving against the current" = Upstream (u - v).

[ TRAP ALERT 3: Relative Length in Passing Man on Train ]
• If Train A passes a man sitting in Train B, DO NOT add Train B's length.
• The distance is purely the length of Train A.
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You

Gemini 3.6 Thinking

D. E. Shaw Quantitative Cheat Sheet

Time & Work: Multi-Worker Mechanics & Rate Variations

To solve complex Time & Work problems at D. E. Shaw speed, completely avoid fractional algebra ($\frac{1}{x} + \frac{1}{y} = \frac{1}{t}$). Instead, standardize all problems using **Unitary Work (LCM Method)** or **Efficiency Ratios**.

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1. Core Principles & Framework

The Master Work Equation

$$\text{Work } (W) = \text{Efficiency } (E) \times \text{Time } (T)$$

- **Efficiency (E):** The amount of work completed per unit time (e.g., units/day).
- **Inverse Relationship:** For a fixed work W, Efficiency and Time are inversely proportional:

$$\frac{E_1}{E_2} = \frac{T_2}{T_1}$$

The LCM Method (4-Step Engine)

Whenever given the individual times taken by workers $A_1, A_2, \ldots, A_n$:

1. **Assume Total Work (W):** Set $W = \text{LCM}(T_1, T_2, \ldots, T_n)$ units.
2. **Calculate Individual Efficiencies:** Compute $E_i = \frac{W}{T_i}$ units/day for each worker.
3. **Combine Rates:** Add or subtract efficiencies based on the scenario:

   $$E_{\text{combined}} = \sum E_{\text{positive}} - \sum E_{\text{negative}}$$

4. **Solve for Target:**

   $$\text{Time} = \frac{\text{Remaining Work}}{E_{\text{active}}}$$

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2. Speed Formulas & High-Frequency Shortcuts

Two and Three Worker Combinations

- **Two Workers (A and B together):**

$$T_{AB} = \frac{A \times B}{A + B}$$

- **Three Workers (A, B, and C together):**

$$T_{ABC} = \frac{A \times B \times C}{A \cdot B + B \cdot C + C \cdot A}$$

The Offset Root Shortcut

Scenario: A and B working together finish a task in T days. Working alone, A takes a days **more** than T, and B takes b days **more** than T.

$T = \sqrt{a \times b}$

> **Example:** If A takes 9 days more than $(A+B)$ and B takes 4 days more than $(A+B)$, then $T = \sqrt{9 \times 4} = 6$ days.

Efficiency Ratios & Percentage Manipulations

- **"A is $k\%$ more efficient than B":**

$\frac{E_A}{E_B} = \frac{100+k}{100} \implies \frac{T_A}{T_B} = \frac{100}{100+k}$

- **"A takes $k\%$ less time than B":**

$\frac{T_A}{T_B} = \frac{100-k}{100} \implies \frac{E_A}{E_B} = \frac{100}{100-k}$

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3. Problem Variations & Step-by-Step Blueprints

Variation 1: Worker Leaves d Days BEFORE Completion

Scenario: A and B start together, but A leaves d days before the work is actually completed.

Blueprint: Virtual Work Addition

Instead of splitting the timeline into unknown variables, force A to stay and add A's missing work to the total target.

1. Calculate total work $W = \text{LCM}(T_A, T_B)$.
2. Find efficiencies E_A and E_B.
3. Add the work A *would have done* in those d days to W:

   $W_{\text{virtual}} = W + (E_A \times d)$
4. Total time taken by B (which is the actual total duration):

   $T_{\text{total}} = \frac{W_{\text{virtual}}}{E_A + E_B}$

Variation 2: Dynamic / Alternate Days Work

Scenario: A and B work on alternate days, starting with A on Day 1, B on Day 2, A on Day 3, etc.

Blueprint: Cycle Bundling

1. Set Total Work $W = \text{LCM}(T_A, T_B)$ and calculate E_A, E_B.
2. Define 1 Cycle = 2 days. Work done in 1 cycle:

   $W_{\text{cycle}} = E_A + E_B$
3. Find full cycles completed:

   $N = \left\lfloor \frac{W}{W_{\text{cycle}}} \right\rfloor$
4. Calculate remaining work: $W_{\text{rem}} = W - (N \times W_{\text{cycle}})$. Time elapsed = $2N$ days.
5. Pass remaining work step-by-step through the rotation order:
   - If $W_{\text{rem}} \leq E_A$: Add $\frac{W_{\text{rem}}}{E_A}$ days.
   - If $W_{\text{rem}} > E_A$: A works 1 full day; remaining $W_{\text{rem}} - E_A$ is finished by B in $\frac{W_{\text{rem}} - E_A}{E_B}$ days.

Variation 3: Group Work & Rate Equivalence (MDH Chain Rule)

Scenario: M_1 men, W_1 women, or C_1 children complete a work in D_1 days working H_1 hours/day.

Master Chain Formula

$$\frac{M_1 \cdot D_1 \cdot H_1 \cdot E_1}{W_1} = \frac{M_2 \cdot D_2 \cdot H_2 \cdot E_2}{W_2}$$

- E = Individual efficiency rate
- W = Total work units completed

Equivalence Conversion Strategy

If given *"x Men or y Women can do a job in D days"*:

1. Equate total daily outputs to find the efficiency ratio:

   $$x \cdot E_M = y \cdot E_W \implies \frac{E_M}{E_W} = \frac{y}{x}$$

2. Substitute $E_M = y$ and $E_W = x$ directly into the target equation.

Variation 4: Wage Distribution Rules

$$\text{Wage Share} \propto \text{Total Work Done by Individual}$$

- **Case A (All workers work for the EXACT same duration):** Wages are divided strictly in the ratio of their **efficiencies**:

  $$\text{Wage}_A : \text{Wage}_B = E_A : E_B$$

- **Case B (Workers work for DIFFERENT durations / leave early):** Wages are divided in the ratio of **actual work completed**:

  $$\text{Wage}_A : \text{Wage}_B = (E_A \times t_A) : (E_B \times t_B)$$

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4. Common Traps & Negative Work Dynamics

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[ TRAP ALERT 1: Alternate Days Final Fraction ]
• Never round up the last day in alternate work.
• Calculate the exact fraction: (Remaining Work / Efficiency of Next Worker).
• Verify WHO starts the work—if B starts instead of A, the remaining work allocation flips.
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[ TRAP ALERT 2: Division of Wages ]
• NEVER divide wages by the ratio of days taken alone (T_A : T_B). Time taken is inversely
  proportional to efficiency.
• Wrong: Divide in ratio 10 : 15.
• Correct: Convert to efficiencies (1/10 : 1/15 = 3 : 2) or use LCM work units.
```

- **Pipes and Negative Work Trap:** When outlet pipes (or destructive workers) are involved, treat their efficiency as negative ($E_{\text{outlet}} = -k$).
  - *Critical Boundary Case:* If a tank is being filled by A and emptied by B on alternate minutes, the tank may reach 100% capacity **during** A's turn before B drains it. Always subtract E_A from total capacity first to find the threshold for safe cycling.

You

Permutations and combinations: word arrangements, selection from groups, circular arrangements

Gemini 3.6 Thinking

D. E. Shaw Quantitative Cheat Sheet

Permutations & Combinations: Words, Groups, & Circular Arrangements

In high-speed test environments, counting problems must be solved using pattern classification rather than manual enumeration. Use the **Block Method**, **Gap Method**, and **Group Factorization** blueprints to solve variations in seconds.

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1. Fundamental Formulas & Identities

Permutations vs. Combinations

- **Permutations (Order Matters):**

  $${}^{n}P_{r} = \frac{n!}{(n-r)!}$$

- **Combinations (Order Does NOT Matter):**

  $${}^{n}C_{r} = \frac{n!}{r!(n-r)!} = \frac{{}^{n}P_{r}}{r!}$$

Essential Combinatorial Identities

- **Symmetry:** ${}^{n}C_{r} = {}^{n}C_{n-r}$
- **Pascal's Identity:** ${}^{n}C_{r} + {}^{n}C_{r-1} = {}^{n+1}C_{r}$
- **Total Subsets:** $\sum_{k=0}^{n} {}^{n}C_{k} = {}^{n}C_{0} + {}^{n}C_{1} + {}^{n}C_{2} + \cdots + {}^{n}C_{n} = 2^{n}$
- **Selecting at least 1 object from n distinct objects:** $2^{n} - 1$

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2. Word Arrangements (Letter Permutations)

Variation 1: Words with Repeating Letters

Scenario: Arrange a word of length N where letter L_1 appears p times, L_2 appears q times, and L_3 appears r times.

$$\text{Total Arrangements} = \frac{N!}{p! \cdot q! \cdot r!}$$

Variation 2: Restricted Grouping — "Always Together" (The Block Method)

Scenario: Arrange letters such that specific letters (e.g., all vowels) must **always stay together**.

Blueprint:

1. Treat the constrained items as **1 single super-item (block)**.
2. Count the total outer items: $(\text{Remaining items} + 1 \text{ block})$.
3. Calculate outer arrangements: $N_{\text{outer}}!$ (adjust for repetitions outside).
4. Calculate inner arrangements inside the block: $N_{\text{inner}}!$ (adjust for repetitions inside).

$$\text{Total Ways} = (\text{Outer Arrangements}) \times (\text{Inner Arrangements})$$

Variation 3: Separation — "Never Together" (The Gap Method)

Scenario: Arrange letters such that specific items (e.g., no two vowels) are **never adjacent**.

Blueprint:

1. Arrange the non-restricted items first in a line. Let there be k such items.
2. Number of arrangements of non-restricted items = $k!$ (account for repetitions).
3. k items create $k+1$ **available gaps** (including the ends).
4. Select m gaps out of $(k+1)$ for the m restricted items: ${}^{k+1}C_m$.
5. Arrange the m restricted items in those selected gaps (account for repetitions).

$$\text{Total Ways} = (\text{Ways to arrange non-restricted}) \times {}^{k+1}C_m \times (\text{Ways to arrange restricted inside gaps})$$

Variation 4: Preserved Order of Specific Letters

Scenario: Arrange the letters of a word such that vowels always appear in their original relative order.

Blueprint:

Treat the items whose relative order is fixed as **identical items**.

$$\text{Total Ways} = \frac{N!}{(\text{Count of fixed-order items})! \times \dots}$$

Variation 5: Dictionary Rank of a Word (No Repetition Shortcut)

Scenario: Find the rank of the word **"EXAM"** in an alphabetical dictionary of its letters.

1. Alphabetical order of letters: **A, E, M, X**
2. Set up the positions for **E - X - A - M**:

Position Letter	Alphabetical Order Remaining	Unused Letters Smaller Than Current	Factorial Weight	Product
E	A, E, M, X	1 (A)	$3! = 6$	$1 \times 6 = 6$
X	A, M, X	2 (A, M)	$2! = 2$	$2 \times 2 = 4$
A	A, M	0	$1! = 1$	$0 \times 1 = 0$
M	M	0	$0! = 1$	$0 \times 1 = 0$

3. $\text{Rank} = \text{Sum of Products} + 1 = (6 + 4 + 0 + 0) + 1 = 11.$

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3. Selection from Groups & Distribution

Variation 1: Group Selection with "At Least / At Most" Constraints

Scenario: Form a committee of K people from M men and W women with a constraint (e.g., "at least 2 women").

- **Direct Method:** Sum valid combinations explicitly:

$$\text{Ways} = \sum_{i=\min}^{\max} \left({}^{W}C_i \times {}^{M}C_{K-i} \right)$$

- **Complement Method (Use when valid cases > invalid cases):**

$$\text{Valid Ways} = \text{Total Unconstrained Combinations} - \text{Invalid Combinations}$$

Variation 2: Division of Objects into Groups

Scenario	Group Sizes	Formula
n **distinct objects into 3 UNEQUAL groups**	a, b, c $(a+b+c=n)$	$\frac{n!}{a!\cdot b!\cdot c!}$
n **distinct objects into** k **EQUAL UNNAMED groups**	g each $(k \cdot g = n)$	$\frac{n!}{(g!)^k \cdot k!}$
n **distinct objects into** k **EQUAL NAMED groups / persons**	g each $(k \cdot g = n)$	$\frac{n!}{(g!)^k}$

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4. Circular Arrangements Mechanics

Standard Circular Permutation Rule

The number of ways to arrange n distinct objects around a fixed circular table is:

$$P_{\text{circular}} = (n-1)!$$

- *Reasoning:* Fixing 1 seat breaks the rotational symmetry, leaving $(n-1)$ positions to arrange linearly.

Clockwise vs. Counter-Clockwise Indistinguishable

When flipping/reversing the circle does not create a new arrangement (e.g., **necklaces, keyrings, garlands**):

$$\text{Ways} = \frac{(n-1)!}{2}$$

Circular Gap Method ("No Two Adjacent" in a Circle)

Scenario: Seat M men and W women around a circular table such that **no two women sit together** $(W \leq M)$.

1. Seat the M men around the circular table first:

$$\text{Ways to seat men} = (M-1)!$$

2. Seating M people in a circle creates **exactly** M **circular gaps** (unlike linear which creates $M+1$).

3. Select W gaps out of M and seat the women:

$$\text{Ways to seat women} = {}^{M}C_W \times W!$$

$$\text{Total Circular Ways} = (M-1)! \times {}^{M}C_W \times W!$$

---

5. Decision Matrix & Common Traps

```
                        [ COUNTING PROBLEM IDENTIFIER ]
                                        |
          +-----------------------------+-----------------------------+
          |                                                           |
  Selection Only?                                              Arrangement?
  (Order doesn't matter)                                      (Order matters)
          |                                                           |
    Use Combinations (nCr)                                +-----------+-----------+
                                                          |                       |
                                                  Linear Setup?           Circular Setup?
                                                          |                       |
                                                   Use n! / nPr            Use (n - 1)!
```

```
[ TRAP ALERT 1: Linear vs. Circular Gap Count ]
• Linear arrangements of N items create (N + 1) gaps.
• Circular arrangements of N items create EXACTLY N gaps.

[ TRAP ALERT 2: Equal Group Division Division Factor ]
• Dividing 12 items into 3 UNNAMED groups of 4: Divide by 3! -> [12! / ((4!)^3 * 3!)]
• Distributing 12 items to 3 NAMED people (4 each): Do NOT divide by 3! -> [12! / (4!)^3]

[ TRAP ALERT 3: Relative Position vs. Absolute Position ]
• If seats around a circular table are NUMBERED (Seat 1, Seat 2, ...), the symmetry
  is broken. Treat as LINEAR arrangement = n! instead of (n - 1)!.
```

You

Probability: basic and conditional probability, sometimes combined with permutations

Gemini 3.6 Thinking

D. E. Shaw Quantitative Cheat Sheet

Probability: Basic, Conditional, & Combinatorial (P&C) Hybrids

D. E. Shaw probability questions often combine **combinatorial counting (P&C)** with **conditional probability** and **Bayesian updating**. The key is setting up the correct sample space quickly without miscalculating dependencies.

---

1. Core Principles & Axioms

Classical Definition

For a sample space S with equally likely outcomes, the probability of event E is:

$$P(E) = \frac{n(E)}{n(S)} = \frac{\text{Favorable Outcomes}}{\text{Total Possible Outcomes}}$$

Fundamental Laws

- **Complement Rule:** $P(A') = 1 - P(A)$
- **Addition Rule (2 Events):**

  $$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

- **Addition Rule (3 Events):**

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C)$$

---

2. Independence vs. Mutual Exclusivity

Property	Mutually Exclusive (Disjoint)	Independent Events
Definition	Events A and B **cannot** occur together.	Occurrence of A gives **zero information** about B.
Intersection	$P(A \cap B) = 0$	$P(A \cap B) = P(A) \cdot P(B)$
Union	$P(A \cup B) = P(A) + P(B)$	$P(A \cup B) = P(A) + P(B) - P(A) \cdot P(B)$
Conditional	$P(A | B) = 0$	

> **Critical Trap:** If $P(A) > 0$ and $P(B) > 0$, mutually exclusive events **can never be independent**. If one happens, the probability of the other drops to zero.

---

3. Conditional Probability & Bayes' Theorem

Conditional Probability Formula

The probability that event A occurs given that event B has already occurred:

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad \text{where } P(B) > 0$$

- **Sample Space Reduction:** Conditioning on B effectively reduces your total sample space from S down to B.

Law of Total Probability

If $B_1, B_2, \ldots, B_n$ form a complete partition of the sample space S (mutually exclusive and exhaustive):

$$P(A) = \sum_{i=1}^{n} P(A|B_i) \cdot P(B_i)$$

Bayes' Theorem Equation

To reverse conditional probabilities (find $P(B_k|A)$ given $P(A|B_k)$):

$$P(B_k|A) = \frac{P(A|B_k) \cdot P(B_k)}{\sum_{i=1}^{n} P(A|B_i) \cdot P(B_i)}$$

Rapid 2x2 Matrix Method for Bayes

For diagnostic/false positive problems, skip full formulas and set up a frequency table based on a hypothetically large population (e.g., $10,000$):

	Event B Happens	Event B Doesn't Happen	Total
Test Positive (A)	True Positives (TP)	False Positives (FP)	$TP + FP$
Test Negative (A')	False Negatives (FN)	True Negatives (TN)	$FN + TN$

$$P(B|A) = \frac{TP}{TP + FP}$$

---

4. Combinatorial Probability (P&C Hybrids)

When outcomes are constructed via permutations or combinations, $n(E)$ and $n(S)$ must be evaluated using P&C rules.

Variation 1: Selection Probability (Hypergeometric Distribution)

Scenario: Selecting n items from a total of N items containing K defective and $(N-K)$ non-defective items. Probability of getting exactly k defectives:

$$P(X=k)=\frac{\binom{K}{k}\cdot\binom{N-K}{n-k}}{\binom{N}{n}}$$

Variation 2: Permutation-Based Lineup Probability

Scenario: N people (including Person X and Person Y) are arranged randomly in a line. What is the probability that X and Y are separated by exactly k people?

1. **Total Arrangements:** $n(S)=N!$
2. **Favorable Arrangements:**
   - Treat (X, \text{ k people }, Y) as a unit or count available position pairs:
   - Positions for (X,Y) pair can be $(1,k+2),(2,k+3),\ldots,(N-k-1,N)$. Total pairs $=N-k-1$.
   - X and Y can swap seats: $2!$ ways.
   - Remaining $(N-2)$ people can be arranged in $(N-2)!$ ways.

$$P=\frac{2\cdot(N-k-1)\cdot(N-2)!}{N!}=\frac{2(N-k-1)}{N(N-1)}$$

Variation 3: Derangement Probability

Scenario: n letters are randomly placed into n addressed envelopes. What is the probability that **no letter goes into its correct envelope**?

- Number of Derangements (D_n):

  $$D_n=n!\sum_{k=0}^{n}\frac{(-1)^k}{k!}=n!\left(1-\frac{1}{1!}+\frac{1}{2!}-\frac{1}{3!}+\cdots+\frac{(-1)^n}{n!}\right)$$

- **Derangement Probability:**

  $$P(\text{Complete Derangement})=\frac{D_n}{n!}\approx\frac{1}{e}\approx 0.3679\quad(\text{for } n\geq 4)$$

Variation 4: Binomial Probability Model

Scenario: n independent trials, each with success probability p and failure probability $q=1-p$. Probability of exactly k successes:

$$P(X=k)=\binom{n}{k}p^k\,q^{n-k}$$

- **At least 1 success in n trials:**

  $$P(X\geq 1)=1-(1-p)^n$$

---

5. Rapid Problem-Solving Blueprints

Blueprint A: "At Least One" Trap

Whenever a problem asks for $P(\text{at least one event occurs})$, turn it inside out immediately:

$$P(\text{At least 1})=1-P(\text{None occur})$$

> **Example:** Rolling a fair die 4 times. Find $P(\text{at least one } 6)$.

> $P(\text{No 6 in one roll}) = \frac{5}{6} \implies P(\text{No 6 in 4 rolls}) = \left(\frac{5}{6}\right)^4$
>
> $P(\text{At least one 6}) = 1 - \left(\frac{5}{6}\right)^4$

Blueprint B: Consecutive Draw Probabilities (With vs. Without Replacement)

- **With Replacement:** Draws are independent. Probabilities remain constant across draws.
- **Without Replacement:** Draws are dependent. Update the numerator and denominator after each draw:

$$P(A_1 \cap A_2 \cap A_3) = P(A_1) \cdot P(A_2|A_1) \cdot P(A_3|A_1 \cap A_2)$$

---

6. Common Traps & Boundary Cases

```
[ TRAP ALERT 1: Conditioning Variable Swap ]
• P(A|B) IS NOT EQUAL TO P(B|A).
• P(A|B) asks: "Out of the times B happened, how often did A happen?"
• P(B|A) asks: "Out of the times A happened, how often did B happen?"
```

```
[ TRAP ALERT 2: Unfair Sample Space ]
• Classical P = n(E)/n(S) ONLY holds if all outcomes in S are EQUALLY LIKELY.
• Example: Tossing 2 coins yields {2 Heads, 1 Head/1 Tail, 2 Tails}.
  These 3 outcomes are NOT equally likely. Expand to {(H,H), (H,T), (T,H), (T,T)}.
```

- **The Birthday Problem Threshold:** In a room of n people, the probability that at least two share a birthday:

  $$P(\text{Shared Birthday}) = 1 - \frac{365 \times 364 \times \cdots \times (365-n+1)}{365^n}$$

  - For $n = 23$, $P > 0.50$ (50%).
  - For $n = 50$, $P > 0.97$ (97%).

You

Data interpretation: bar charts, tables, pie charts — inference questions, not just reading values

Gemini 3.6 Thinking

D. E. Shaw Quantitative Cheat Sheet

Data Interpretation: Advanced Inference, Charts, & Analytical Mechanics

D. E. Shaw's Data Interpretation (DI) section focuses heavily on **data sufficiency, logical inference, mix shifts, and growth trends**, rather than straightforward value reading. Speed depends on rapidly verifying whether statements are mathematically proven, possible, or impossible without doing full calculations.

---

1. Core Mathematical Metrics & Growth Formulas

Growth Rates: Simple, Average, & Compound

- **Absolute Growth:**

$$\Delta V = V_{\text{final}} - V_{\text{initial}}$$

- **Simple Percentage Growth:**

$$\%\,\text{Growth} = \left(\frac{V_{\text{final}} - V_{\text{initial}}}{V_{\text{initial}}}\right) \times 100$$

- **Average Annual Growth Rate (AAGR):** Arithmetic mean of yearly growth rates over n periods.

$$\text{AAGR} = \frac{g_1 + g_2 + \cdots + g_n}{n}$$

- **Compound Annual Growth Rate (CAGR):** The geometric growth rate over n years.

$$\text{CAGR} = \left(\frac{V_n}{V_0}\right)^{\frac{1}{n}} - 1$$

The CAGR Approximation Rule

To avoid calculating fractional powers without a scientific calculator, approximate CAGR using AAGR and a correction factor:

$$\text{CAGR} \approx \text{AAGR} - \frac{\text{Variance of Growth Rates}}{2}$$

> **Rule of Thumb:**
>
> - If annual growth rates are relatively constant ($g_1 \approx g_2 \approx g_3$), $\text{CAGR} \approx \text{AAGR}$.
> - If annual growth rates fluctuate wildly, $\text{CAGR} < \text{AAGR}$. The greater the volatility, the larger the gap.

The Mix Shift / Weighted Average Equation

When an aggregate metric depends on sub-components with individual values (A_i) and weights (w_i):

$$A_{\text{overall}} = \frac{w_1 A_1 + w_2 A_2 + \cdots + w_k A_k}{w_1 + w_2 + \cdots + w_k} = \sum_{i=1}^{k} (\text{Share}_i \times A_i)$$

---

2. Rapid Fraction & Ratio Comparison Hacks

In multi-column tables or complex charts, comparing fractions quickly is critical.

Technique 1: Cross-Multiplication (Exact for 2 Fractions)

To compare $\frac{a}{b}$ and $\frac{c}{d}$:

$$\text{Calculate } a \cdot d \quad \text{and} \quad b \cdot c$$

- If $a \cdot d > b \cdot c \implies \frac{a}{b} > \frac{c}{d}$

Technique 2: Percentage Change in Numerator vs. Denominator

To compare $\frac{N_1}{D_1}$ and $\frac{N_2}{D_2}$:

Calculate $\%\Delta N = \frac{N_2 - N_1}{N_1}$ and $\%\Delta D = \frac{D_2 - D_1}{D_1}$.

- If $\%\Delta N > \%\Delta D \implies \frac{N_2}{D_2} > \frac{N_1}{D_1}$ (Fraction increased)
- If $\%\Delta N < \%\Delta D \implies \frac{N_2}{D_2} < \frac{N_1}{D_1}$ (Fraction decreased)

Technique 3: The Benchmark Interpolation Method

Compare arbitrary fractions against known anchor values ($0.25, 0.33, 0.50, 0.66, 0.75$):

$$\text{Is } \frac{47}{91} > \frac{53}{112}?$$

- $\frac{47}{91} > \frac{45.5}{91} = 0.50$ (Greater than 50%)
- $\frac{53}{112} < \frac{56}{112} = 0.50$ (Less than 50%)
- Conclusion: $\frac{47}{91} > \frac{53}{112}$ instantly without division.

---

3. Chart Types & Inference Patterns

1. Pie Charts: Single vs. Multi-Pie Dynamics

- **Degree-to-Percentage Conversion:**

$$\text{Percentage } (\%) = \frac{\text{Angle } (\theta)}{3.6} \qquad \text{Angle } (\theta) = \text{Percentage } (\%) \times 3.6$$

- **Dual Pie Chart Base Trap:**

```
Pie Chart A (Year 1)              Pie Chart B (Year 2)
Total Sales = $10 Million          Total Sales = $50 Million
Segment X = 40% ($4M)              Segment X = 20% ($10M)
```

> **Key Inference Rule:** A **decrease in percentage share** between two pie charts DOES NOT mean a **decrease in absolute value** unless the total base remains constant or decreases. In the example above, Segment X's share dropped from 40% to 20%, but its absolute value increased by 150% ($4\text{M} \rightarrow 10\text{M}$).

---

2. Stacked Bar Charts vs. 100% Stacked Bar Charts

Chart Type	Primary Data Given	Valid Inferences	Invalid Inferences Without External Data
Standard Stacked Bar	Absolute component values & total heights	• Total absolute growth • Absolute component growth • Relative component shares	• None (all absolute metrics are present)
100% Stacked Bar	Percentage shares of components (100% scale)	• Market share trends • Proportional composition shifts	• Absolute magnitude of any component • Total overall growth trends

---

3. Missing Data Tables

When presented with incomplete tables:

1. **Row/Column Integrity:** Use $\text{Total} = \sum \text{Components}$ to fill missing entries first.
2. **Ratio Integrity:** Check if marginal totals match cross-sectional percentages.
3. **Degree of Freedom Check:** If k entries are missing in a system of n independent linear equations:
   - If $k \leq n$: Missing values can be uniquely determined (Definite Inference).
   - If $k > n$: Range/Bounds can be established, but exact values **cannot** be inferred.

---

4. Logical Inference Frameworks & Fallacies

Simpson's Paradox (The Mix Shift Trap)

An overall aggregate percentage/average can move in the **opposite direction** of every individual sub-group if the underlying weights shift dramatically.

Example Scenario:

- **Department A Success Rate:** Increases from 60% to 70%.
- **Department B Success Rate:** Increases from 20% to 30%.
- **Overall Company Success Rate:** Can still **DROP** if the proportion of applications shifts heavily toward Department B (the lower-performing department).

Inference Checklist: Never evaluate aggregate growth without checking component weight shifts.

Relative vs. Absolute Data Sufficiency Rules

```
                    [ EVALUATING AN INFERENCE STATEMENT ]
                                      |
      +-------------------------------+-------------------------------+
      |                                                               |
Is the question asking for ABSOLUTE values?             Is the question asking for RELATIVE values?
 (e.g., "$ Amount", "Units", "People")                    (e.g., "% Share", "Ratio", "% Change")
      |                                                               |
      +---> REQUIRES at least one absolute                  +---> REQUIRES only proportional data.
            baseline metric in the chart.                         Absolute totals are NOT needed.
```

---

5. Common Traps & Precision Checklist

```
[ TRAP ALERT 1: Graph Axis Distortions ]
• Non-Zero Origin (Truncated Y-Axis): A bar starting at 90 instead of 0 makes 95 look
  double the size of 91. Always read axis tick values, do not judge visually.
• Logarithmic Axes: Constant visual steps represent exponential growth (10, 100, 1000).

[ TRAP ALERT 2: "Cannot Be Determined" Options ]
• Do not pick "Cannot be determined" simply because the exact value takes time to calculate.
• Pick "Cannot be determined" ONLY if an essential variable (e.g., the total base value
  in percentage charts) is completely absent.

[ TRAP ALERT 3: Average of Averages ]
• Avg(Overall) IS NOT equal to (Avg1 + Avg2 + Avg3) / 3 unless the sample sizes
  n1 = n2 = n3 are identical. Always apply Weighted Averages.
```

You

Number systems: HCF, LCM, and divisibility rules

Gemini 3.6 Thinking

D. E. Shaw Quantitative Cheat Sheet

Number Systems: HCF, LCM, & Divisibility Mechanics

Speed in Number Systems comes from recognizing algebraic structures behind divisibility and using modular arithmetic shortcuts instead of brute-force division.

---

1. Core Theorems & Multi-Number Relations

Prime Factorization Representation

Any composite integer $N > 1$ can be uniquely expressed as:

$$N = p_1^a \cdot p_2^b \cdot p_3^c \cdots p_k^k$$

- $\text{HCF}(A, B)$: Product of the **lowest** powers of common prime factors.
- $\text{LCM}(A, B)$: Product of the **highest** powers of all prime factors.

Key HCF-LCM Identities

$$\text{HCF}(A, B) \times \text{LCM}(A, B) = A \times B \quad (\text{Valid ONLY for 2 numbers})$$

For three numbers A, B, and C:

$$\text{LCM}(A, B, C) = \frac{A \cdot B \cdot C \cdot \text{HCF}(A,B,C)}{\text{HCF}(A,B) \cdot \text{HCF}(B,C) \cdot \text{HCF}(C,A)}$$

$$\text{HCF}(A, B, C) \times \text{LCM}(A, B, C) \neq A \times B \times C \quad (\text{Do NOT use 2-number rule})$$

HCF & LCM of Fractions

$$\text{HCF}\left(\frac{a}{b}, \frac{c}{d}, \frac{e}{f}\right) = \frac{\text{HCF}(a,c,e)}{\text{LCM}(b,d,f)}$$

$$\text{LCM}\left(\frac{a}{b}, \frac{c}{d}, \frac{e}{f}\right) = \frac{\text{LCM}(a,c,e)}{\text{HCF}(b,d,f)}$$

> **Critical Pre-requisite:** All fractions **must** be reduced to their lowest terms (simplified forms) before applying these formulas.

---

2. Divisibility Rules & Oscillator Engine

Standard Divisibility Matrix

Divisor	Condition / Test Rule	Example / Shortcut
2^k $(2, 4, 8, 16)$	Last k digits must be divisible by 2^k.	For 8 (2^3), check last 3 digits.
5^k $(5, 25, 125)$	Last k digits must be divisible by 5^k.	For 25 (5^2), last 2 digits must be 00, 25, 50, or 75.
3 and 9	Sum of all digits must be divisible by 3 or 9.	Digital Root Method.
11	Difference between sum of odd-positioned digits and even-positioned digits is 0 or a multiple of 11.	$
Composite $N = a \cdot b$	Divisible by BOTH a and b, where $\gcd(a, b) = 1$.	For 72, check 8 and 9 (Not 6 and 12).

Prime Oscillators (Universal Rule for Odd Primes)

To test if a number N is divisible by a prime P, isolate the last digit d, multiply it by the **Oscillator Key (K)**, and add/subtract from the remaining truncated number R:

$$N' = R \pm (d \times K)$$

Prime (P)	Oscillator Formula	Key (K)	Operation
7	$R - 2d$	2	Subtract
13	$R + 4d$	4	Add
17	$R - 5d$	5	Subtract
19	$R + 2d$	2	Add
23	$R + 7d$	7	Add
29	$R + 3d$	3	Add

---

3. Standard Remainder Variations & Blueprints

Category A: LCM-Based Problems (Finding Dividend)

Variation 1: Same Positive Remainder (r)

Scenario: Find the smallest number which when divided by a, b, c leaves the same remainder r in each case.

$$N = \text{LCM}(a, b, c) \cdot k + r \quad (k \geq 0)$$

Variation 2: Constant Difference Remainder (K)

Scenario: Find the smallest number which when divided by a, b, c leaves remainders r_1, r_2, r_3 respectively, such that $(a - r_1) = (b - r_2) = (c - r_3) = K$.

$$N = \text{LCM}(a, b, c) \cdot k - K$$

Category B: HCF-Based Problems (Finding Divisor)

Variation 3: Explicit Known Remainders

Scenario: Find the largest number that divides x, y, z leaving remainders r_1, r_2, r_3 respectively.

$$\text{Required Divisor} = \text{HCF}(x - r_1,\ y - r_2,\ z - r_3)$$

Variation 4: Same Unknown Remainder (r)

Scenario: Find the largest number that divides x, y, z leaving the **same remainder** in each case (value of r is not given).

$$\text{Required Divisor} = \text{HCF}(|x - y|,\ |y - z|,\ |z - x|)$$

Category C: Coprime Pair Systems

Scenario: Given $\text{HCF}(A, B) = H$ and $\text{LCM}(A, B) = L$. Find the number of possible pairs (A, B).

1. Express numbers in terms of HCF:

   $$A = H \cdot a \quad \text{and} \quad B = H \cdot b \quad \text{where } \gcd(a, b) = 1$$

2. Substitute into LCM relation:

   $$L = H \cdot a \cdot b \implies a \cdot b = \frac{L}{H}$$

3. Find all coprime factor pairs (a, b) of the value $\frac{L}{H}$. Number of valid pairs of (A, B) equals the number of co-prime pairs (a, b).

---

4. Advanced Modular Remainder Theorems

1. Fermat's Little Theorem

If p is a prime number and $\gcd(a, p) = 1$:

$$a^{p-1} \equiv 1 \pmod{p} \implies \text{Remainder}\left(\frac{a^{p-1}}{p}\right) = 1$$

---

2. Euler's Totient Theorem

If $\gcd(a, N) = 1$:

$$a^{\phi(N)} \equiv 1 \pmod{N}$$

- **Totient Function $\phi(N)$:** Count of numbers up to N coprime to N.

$$\phi(N) = N\left(1 - \frac{1}{p_1}\right)\left(1 - \frac{1}{p_2}\right)\dots\left(1 - \frac{1}{p_k}\right)$$

> **Example:** Find remainder of $7^{82} \pmod{100}$. $\phi(100) = 100\left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{5}\right) = 40$. $7^{40} \equiv 1 \pmod{100} \implies 7^{82} = (7^{40})^2 \cdot 7^2 \equiv 1^2 \cdot 49 \equiv 49 \pmod{100}$.

---

3. Wilson's Theorem

For any prime p:

$$(p-1)! \equiv -1 \pmod{p} \quad \text{or} \quad (p-1)! + 1 \text{ is divisible by } p$$

$$(p-2)! \equiv 1 \pmod{p}$$

---

5. Common Traps & Boundary Cases

```
[ TRAP ALERT 1: Un-simplified Fraction HCF/LCM ]
• Wrong: Find HCF of 2/4 and 3/6 using formulas directly.
• Correct: Reduce to lowest terms first (1/2 and 1/2), then evaluate.
```

```
[ TRAP ALERT 2: "Smallest N-Digit Number" Bounds ]
• For "Smallest 4-digit number satisfying condition N = LCM*k + r":
  1. Find baseline LCM.
  2. Divide 1000 by LCM to find minimum integer multiplier k.
  3. Calculate (LCM * k) + r. Do NOT just add r to 1000!
```

- **Negative Remainder Shortcut:** When evaluating $\frac{17^{29}}{18}$, rewrite 17 as $(-1) \pmod{18}$:

$$\text{Remainder} = (-1)^{29} \pmod{18} = -1 \equiv 18 - 1 = 17$$

You

Mixtures and allegations: alligation method, three-component blending problems

Gemini 3.6 Thinking

D. E. Shaw Quantitative Cheat Sheet

Mixtures & Alligations: Dual & Triple Component Blending Mechanics

The Rule of Alligation is an optimized graphical shortcut for the **Weighted Average** formula. In fast-paced tests, alligation allows you to skip algebraic multi-variable equations for liquid mixtures, profit/loss splits, interest rates, and multi-component blending.

---

1. Core Mathematical Foundation & The Alligation Cross

Weighted Average Linkage

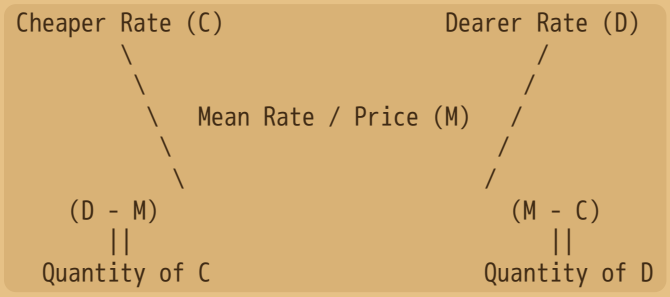
If two ingredients of unit values (or concentrations) C (Cheaper) and D (Dearer) are mixed in quantities Q_c and Q_d to produce a mixture of mean value M:

$$M = \frac{Q_c \cdot C + Q_d \cdot D}{Q_c + Q_d}$$

Rearranging gives the **Fundamental Equation of Alligation**:

$$\frac{Q_c}{Q_d} = \frac{D - M}{M - C}$$

The Alligation Cross Diagram



$$\text{Ratio of Quantities } (Q_c : Q_d) = (D - M) : (M - C)$$

> **Golden Rule of Alligation Values:**
>
> - The mean value M must **always** lie strictly between C and D: $C < M < D$.
> - C, D, and M must all be defined with respect to the **same base unit** (e.g., all cost prices, or all percentage concentrations of the *same* solute).

---

2. Removal and Replacement Dynamics (Successive Dilution)

Scenario: A container initially holds V units of pure liquid A. A volume x is drawn off and replaced with an equal volume of liquid B (water/solvent). This process is repeated n times in total.

Master Dilution Formula

The quantity of original liquid A remaining in the container after n operations:

$$Q_A = V \times \left(1 - \frac{x}{V}\right)^n$$

The concentration of liquid A after n operations:

$$\text{Concentration } (\%) = \left(1 - \frac{x}{V}\right)^n \times 100$$

Non-Uniform Removal Volumes

If different amounts $x_1, x_2, \ldots, x_n$ are removed and replaced with liquid B sequentially:

$$Q_A = V \times \left(1 - \frac{x_1}{V}\right)\left(1 - \frac{x_2}{V}\right)\ldots\left(1 - \frac{x_n}{V}\right)$$

Ratio of Original Liquid to Added Solvent

After n replacements, the ratio of Liquid A to Liquid B inside the vessel is:

$$\frac{Q_A}{Q_B} = \frac{\left(1-\frac{x}{V}\right)^n}{1-\left(1-\frac{x}{V}\right)^n}$$

---

3. Three-Component Blending Problems (Triple Alligation)

When blending three components A, B, C with prices/concentrations $a < b < c$ to achieve a target mean value m, the system has **infinite valid ratio combinations** unless fixed constraints are given. To generate valid integer ratio solutions quickly, use the **Pairing Blueprint**.

Blueprint: The Pairing Rule for 3 Components

Group the components into pairs such that **each pair contains one value less than m and one value greater than m.**

Case 1: Mean lies between middle and highest ($a < b < m < c$)

Here, $a < m$ and $b < m$, while only $c > m$.

- **Pair 1:** (a, c)
- **Pair 2:** (b, c)

1. **Alligate Pair 1 (a, c) for mean m:**

   $$q_{a1} : q_{c1} = (c - m) : (m - a)$$

2. **Alligate Pair 2 (b, c) for mean m:**

   $$q_{b2} : q_{c2} = (c - m) : (m - b)$$

3. **Combine the Quantities:**
   - $Q_a = q_{a1} = (c - m)$
   - $Q_b = q_{b2} = (c - m)$
   - $Q_c = q_{c1} + q_{c2} = (m - a) + (m - b)$

$$\text{Final Ratio } (Q_a : Q_b : Q_c) = (c - m) : (c - m) : [(m - a) + (m - b)]$$

Case 2: Mean lies between lowest and middle ($a < m < b < c$)

Here, only $a < m$, while $b > m$ and $c > m$.

- **Pair 1:** (a, b)
- **Pair 2:** (a, c)

1. **Alligate Pair 1 (a, b):** $q_{a1} : q_{b1} = (b - m) : (m - a)$
2. **Alligate Pair 2 (a, c):** $q_{a2} : q_{c2} = (c - m) : (m - a)$
3. **Combine the Quantities:**
   - $Q_a = q_{a1} + q_{a2} = (b - m) + (c - m)$
   - $Q_b = q_{b1} = (m - a)$
   - $Q_c = q_{c2} = (m - a)$

$$\text{Final Ratio } (Q_a : Q_b : Q_c) = [(b-m)+(c-m)] : (m-a) : (m-a)$$

Deterministic Solution via Linear Algebra System

If the total final quantity Q_{total} and one of the ingredient quantities (e.g., Q_a) are fixed, solve using the exact system of linear equations:

Q_a + Q_b + Q_c = Q_{\text{total}} \\ a \cdot Q_a + b \cdot Q_b + c \cdot Q_c = m \cdot Q_{\text{total}} \end{cases}$$ 1. Substitute known Q_a and Q_{total}. 2. Solve the 2-variable linear system for Q_b and Q_c. --- ## 4. Application Cross-Over Blueprints Alligation applies to any weighted average problem of the form $\text{Rate} = \frac{\text{Numerator}}{\text{Denominator}}$. The resulting ratio always represents the **Denominator quantities**. | Application Area | Cheaper Rate (C) | Dearer Rate (D) | Mean Rate (M) | Calculated Ratio ($Q_c : Q_d$) | | :--- | :--- | :--- | :--- | :--- | | **Profit & Loss** | $\% \text{ Profit}_1$ | $\% \text{ Profit}_2$ | $\% \text{ Overall Profit}$ | **Ratio of Cost Prices ($CP_1 : CP_2$)** | | **Time, Speed, Distance** | Speed V_1 | Speed V_2 | Average Speed V_{avg} | **Ratio of Time Taken ($T_1 : T_2$)** | | **Simple Interest** | Interest Rate $R_1\%$ | Interest Rate $R_2\%$ | Overall Rate $R_{\text{net}}\%$ | **Ratio of Principal Amounts ($P_1 : P_2$)** | | **Demographics / Wages** | Avg Wage (Group 1) | Avg Wage (Group 2) | Avg Wage (Overall) | **Ratio of Headcount ($N_1 : N_2$)** | --- ## 5. Critical Traps & Precision Checklist ``` [TRAP ALERT 1: The Base Variable Denominator Trap] • The ratio obtained from Alligation ALWAYS belongs to the DENOMINATOR of the rate used. • Example: Alligating Speed (km / hour) yields the ratio of TIME (hours), NOT distance! • Example: Alligating Profit % (Profit / CP) yields the ratio of COST PRICES, NOT selling prices! ``` ``` [TRAP ALERT 2: Profit/Loss Sign Inversion] • When alligating transactions involving a loss, assign a NEGATIVE sign to the loss rate. • Example: 20% Profit (+20) mixed with 10% Loss (-10) to make an overall 5% Profit (+5): - Cheaper = -10, Dearer = +20, Mean = +5 - Ratio = (20 - 5) : (5 - (-10)) = 15 : 15 = 1 : 1. ``` * **Selling Price Conversion Trap:** Never run alligation using Selling Prices if the profit percentages on the two items are different. Always convert SP to CP or alligate using Profit % directly: $$CP = \frac{SP}{1 + \frac{P\%}{100}}$$ * **Water Dilution Value Trap:** If pure water is added to a milk/acid mixture: * If using **Milk concentration %**, Water has **0% concentration**. * If using **Water concentration %**, Water has **100% concentration**. * If using **Cost Price**, Water is assumed to cost **$\$0$ per liter** unless specified otherwise.

You

Ratios and proportions: direct and inverse proportion under varied conditions

Gemini 3.6 Thinking

D. E. Shaw Quantitative Cheat Sheet

Ratios & Proportions: Direct, Inverse, & Joint Proportionality

In high-speed test environments, standard algebraic substitution often leads to redundant calculations. Master **Joint Multiplier Algebra**, **Ratio Bridging**, and the **Cross-Multiplication Blueprint** to solve complex variation problems in seconds.

---

1. Core Principles & Master Equations

Fundamental Definitions

- **Direct Proportion ($y \propto x$):** As x increases, y increases proportionally.

  $$y = k \cdot x \implies \frac{y_1}{x_1} = \frac{y_2}{x_2} = k$$

- **Inverse Proportion ($y \propto \frac{1}{x}$):** As x increases, y decreases proportionally.

  $$y \cdot x = k \implies y_1 \cdot x_1 = y_2 \cdot x_2 = k$$

- **Joint / Compound Proportion ($y \propto \frac{x^a \cdot z^b}{w^c}$):** Combining multiple direct and inverse dependencies into a single master equation.

  $$\frac{y \cdot w^c}{x^a \cdot z^b} = k \implies \frac{y_1 \cdot w_1^c}{x_1^a \cdot z_1^b} = \frac{y_2 \cdot w_2^c}{x_2^a \cdot z_2^b}$$

Ratio Manipulations & Componendo-Dividendo

If $\frac{a}{b} = \frac{c}{d}$, then the following structural properties hold:

- **Componendo & Dividendo:**

  $$\frac{a+b}{a-b} = \frac{c+d}{c-d}$$

- **Equal Ratios Theorem (Invertendo / Addendo):**

  $$\frac{a}{b} = \frac{c}{d} = \frac{e}{f} = \frac{p \cdot a + q \cdot c + r \cdot e}{p \cdot b + q \cdot d + r \cdot f}$$

- **Power Ratios:**
  - **Duplicate Ratio:** $a^2 : b^2$
  - **Triplicate Ratio:** $a^3 : b^3$
  - **Sub-duplicate Ratio:** $\sqrt{a} : \sqrt{b}$
  - **Sub-triplicate Ratio:** $\sqrt[3]{a} : \sqrt[3]{b}$

---

2. Problem Variations & Rapid Blueprints

Variation 1: Non-Linear Joint Variation (e.g., Broken Value Problems)

Scenario: *"The value of a diamond is directly proportional to the square of its weight ($V \propto W^2$). A diamond weighing W breaks into 3 pieces with weight ratios $a : b : c$. Find the percentage loss in value."*

Blueprint:

1. Set initial weight $W_{\text{total}} = a + b + c$.
2. Initial Value: $V_{\text{initial}} = k \cdot (a + b + c)^2$.
3. New Value (sum of broken pieces): $V_{\text{new}} = k \cdot (a^2 + b^2 + c^2)$.
4. Absolute Value Loss:

   $$\Delta V = V_{\text{initial}} - V_{\text{new}} = 2k(ab + bc + ca)$$

5. **Percentage Loss in Value:**

   $$\%\,\text{Loss} = \left(\frac{2(ab+bc+ca)}{(a+b+c)^2}\right) \times 100$$

Variation 2: Percentage Changes under Joint Conditions

Scenario: *"Variable Z varies directly as X^2 and inversely as $\sqrt{Y}$. If X increases by 20% and Y decreases by 36%, find the percentage change in Z."*

Blueprint: Multiplier Algebra

Convert each percentage change into a growth multiplier factor (K):

- Multiplier for X: $K_X = 1 + 0.20 = 1.2 = \frac{6}{5}$
- Multiplier for Y: $K_Y = 1 - 0.36 = 0.64 = \frac{16}{25}$

Apply the power relations directly to find the net multiplier K_Z:

$$K_Z = \frac{(K_X)^2}{\sqrt{K_Y}} = \frac{\left(\frac{6}{5}\right)^2}{\sqrt{\frac{16}{25}}} = \frac{\frac{36}{25}}{\frac{4}{5}} = \frac{36}{25} \times \frac{5}{4} = \frac{9}{5} = 1.80$$

$$\% \, \Delta Z = (K_Z - 1) \times 100 = (1.80 - 1) \times 100 = +80\% \text{ (Increase)}$$

Variation 3: Composite Additive Proportionality (Fixed + Variable Costs)

Scenario: *"The total expense of an event is partly fixed and partly varies directly with the number of guests. If 100 guests cost \5,000 and 200 guests cost $8,000$, find the cost for 300 guests."*

Blueprint: Linear System Setup

Set the total equation: $E = F + (k \cdot N)$, where F is fixed cost and k is variable rate per guest.

1. Write two simultaneous equations:

   $$E_1 = F + 100k = 5000$$

   $$E_2 = F + 200k = 8000$$

2. Subtract to find k:

   $$(200 - 100)k = 8000 - 5000 \implies 100k = 3000 \implies k = 30$$

3. Substitute k to find F: $F = 5000 - 100(30) = 2000$.

4. Calculate target: E_3 = 2000 + 300(30) = \11,000$.

Variation 4: Income, Expenditure, & Savings Systems

Scenario: *"Incomes of A and B are in ratio $a : b$. Their expenditures are in ratio $c : d$. A saves \S_A and B saves S_B. Find the income of A."*

Blueprint: Cross-Multiplication Method

Set up the matrix of ratios and absolute savings values:

```
Incomes:       a  :  b
Expenditures:  c  :  d
Savings:      SA  :  SB
```

1. Formulate the multiplier variable x:

   $$x = \frac{|d \cdot S_A - c \cdot S_B|}{|a \cdot d - b \cdot c|}$$

2. Incomes are $A = a \cdot x$ and $B = b \cdot x$.

---

3. Ratio Bridging & Scaling Shortcuts

Multi-Variable Ratio Unification

To combine ratios $A : B = a : b$, $B : C = c : d$, and $C : D = e : f$ into $A : B : C : D$:

```
A   :   B   :   C   :   D
a   :   b
        c   :   d
                e   :   f
------------------------
(a·c·e) : (b·c·e) : (b·d·e) : (b·d·f)
```

Unified Ratio $A : B : C : D = (a \cdot c \cdot e) : (b \cdot c \cdot e) : (b \cdot d \cdot e) : (b \cdot d \cdot f)$

---

4. Common Traps & Precision Checklist

```
[ TRAP ALERT 1: Non-Linear Power Inversion ]
• "Y is inversely proportional to the SQUARE of X" means Y ∝ 1 / X².
• Do NOT write Y ∝ 1 / X. Always match exponents precisely.
• Doubling X reduces Y by a factor of 4, not 2.
```

```
[ TRAP ALERT 2: Additive Shifts in Ratios ]
• If A : B = 3 : 4, adding equal absolute amounts (+k) to both numbers
  DOES NOT maintain the ratio.
• (3 + k) / (4 + k) ≠ 3/4.
• Adding a positive quantity to both terms brings the ratio closer to 1.
```

- **Lack of Absolute Anchor Trap:** If a question gives *only* relative ratios (e.g., ratio of incomes, ratio of expenditures, ratio of savings) without giving at least **one absolute dollar/unit value**, absolute values **cannot be determined**. Pick *"Cannot be determined"* instantly.

You

Sequences and series: arithmetic, geometric, and mixed-ratio progressions

Gemini 3.6 Thinking

D. E. Shaw Quantitative Cheat Sheet

Sequences & Series: Arithmetic, Geometric, & Mixed Progressions

Questions on sequences and series in high-speed quantitative tests require rapid pattern identification. Master the **Shift-and-Subtract Blueprint** for mixed progressions, **Telescoping Decompositions**, and **AM-GM-HM Bounds** to avoid tedious term-by-term calculations.

---

1. Core Formulas & Master Identities

Arithmetic Progression (AP)

For an AP with first term a and common difference d:

- **n-th Term (a_n):**

  $a_n = a + (n-1)d$

- **Sum of n Terms (S_n):**

  $S_n = \frac{n}{2}[2a + (n-1)d] = \frac{n}{2}(a + l)$

(where $l = a_n$ is the last term)

- **Arithmetic Mean (AM):** For n terms, $\text{AM} = \frac{a_1+a_2+\cdots+a_n}{n}$.

Geometric Progression (GP)

For a GP with first term a and common ratio r:

- **n-th Term (a_n):**

  $a_n = a \cdot r^{n-1}$

- **Finite Sum (S_n):**

  $S_n = a \cdot \frac{r^n-1}{r-1} \quad (r \neq 1)$

- **Infinite Sum (S_∞):** Valid **only** when $|r| < 1$:

  $S_\infty = \frac{a}{1-r}$

- **Geometric Mean (GM):** For n positive terms, $\text{GM} = (a_1 \cdot a_2 \ldots a_n)^{\frac{1}{n}}$.

Special Standard Series Sums

- **Sum of first n natural numbers:**

  $\sum_{k=1}^{n} k = \frac{n(n+1)}{2}$

- **Sum of squares:**

  $\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$

- **Sum of cubes:**

  $\sum_{k=1}^{n} k^3 = \left[\frac{n(n+1)}{2}\right]^2 = \left(\sum_{k=1}^{n} k\right)^2$

---

2. Arithmetico-Geometric Progression (AGP)

An AGP is formed by multiplying corresponding terms of an AP and a GP:

$$\text{AGP Series: } a,\ (a+d)r,\ (a+2d)r^2,\ (a+3d)r^3,\ \ldots,\ [a+(n-1)d]r^{n-1}$$

The Universal Shift-and-Subtract Blueprint (Finite AGP)

To find the sum S_n of a finite AGP:

1. Write the sum equation explicitly:

   $S_n = a + (a+d)r + (a+2d)r^2 + \cdots + [a+(n-1)d]r^{n-1}$

2. Multiply the entire equation by the common ratio r and shift terms right by 1 position:

   $rS_n = ar + (a+d)r^2 + \cdots + [a+(n-2)d]r^{n-1} + [a+(n-1)d]r^n$

3. Subtract the second equation from the first:

   $(1-r)S_n = a + d(r + r^2 + r^3 + \cdots + r^{n-1}) - [a+(n-1)d]r^n$

4. Simplify the inner GP sum and divide by $(1-r)$.

Direct Infinite AGP Formula

When $|r| < 1$, as $n \to \infty$, the trailing term vanishes. Use this direct shortcut:

$$S_\infty = \frac{a}{1-r} + \frac{d \cdot r}{(1-r)^2}$$

> **Example:** Find $S_\infty = 1 + \frac{4}{5} + \frac{7}{25} + \frac{10}{125} + \dots$
>
> - $a = 1, d = 3, r = \frac{1}{5}$
> - $S_\infty = \frac{1}{1-1/5} + \frac{3 \cdot (1/5)}{(1-1/5)^2} = \frac{5}{4} + \frac{3/5}{16/25} = \frac{5}{4} + \frac{15}{16} = \frac{35}{16}$

---

3. Advanced Series Techniques

Technique 1: Telescoping Series (Method of Differences)

When the general term a_n can be expressed as a difference of consecutive functions: $a_n = V_n - V_{n-1}$.

$$\sum_{k=1}^{n} a_k = (V_1 - V_0) + (V_2 - V_1) + \dots + (V_n - V_{n-1}) = V_n - V_0$$

Common Partial Fraction Templates:

- **Two-Factor Denominators:**

  $$\frac{1}{n(n+k)} = \frac{1}{k}\left(\frac{1}{n} - \frac{1}{n+k}\right)$$

- **Three-Factor Denominators:**

  $$\frac{1}{n(n+1)(n+2)} = \frac{1}{2}\left[\frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)}\right]$$

Technique 2: AM-GM-HM Inequality Bounds

For any n **positive real numbers** $x_1, x_2, \dots, x_n$:

$$\text{AM} \geq \text{GM} \geq \text{HM}$$

$$\frac{x_1 + x_2 + \dots + x_n}{n} \geq (x_1 \cdot x_2 \dots x_n)^{\frac{1}{n}} \geq \frac{n}{\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_n}}$$

- **Equality Condition:** $\text{AM} = \text{GM} = \text{HM}$ holds **if and only if** $x_1 = x_2 = \dots = x_n$.
- **Use Cases:** Finding minimum values of expressions like $x + \frac{k}{x}$ or maximizing products under fixed sums.

---

4. Rapid Variable Selection & Quadratic Properties

Strategic Term Selection for Symmetry

Target Problem Scenario	Recommended Variables	Advantage
3 terms in AP (Sum is given)	$a - d,\ a,\ a + d$	Sum $= 3a$ (Isolates a instantly)
4 terms in AP (Sum is given)	$a - 3d,\ a - d,\ a + d,\ a + 3d$	Common diff is $2d$; Sum $= 4a$
3 terms in GP (Product is given)	$\frac{a}{r},\ a,\ a \cdot r$	Product $= a^3$ (Isolates a instantly)

The AP Quadratic Sum Property

If the sum of n terms of a sequence is represented by a quadratic polynomial in n:

$$S_n = An^2 + Bn$$

Then the sequence is **guaranteed to be an AP** with:

- **Common Difference (d):** $d = 2A$
- **First Term (a_1):** $a_1 = A + B$
- **n-th Term (a_n):** $a_n = S_n - S_{n-1} = 2An + (B - A)$

---

5. Common Traps & Precision Checklist

```
[ TRAP ALERT 1: Infinite GP Convergence Condition ]
• S_infinity = a / (1 - r) ONLY works if |r| < 1.
• If r >= 1 or r <= -1, the infinite sum diverges (infinity / undefined).
```

```
[ TRAP ALERT 2: AM-GM Positivity Constraint ]
• AM >= GM CANNOT be directly applied if any term is negative or zero.
• Terms MUST be strictly positive real numbers (x_i > 0).
```

- **AGP Off-By-One Exponent Trap:** In the n-th term of an AGP, $[a + (n-1)d]r^{n-1}$, make sure the power of r matches the term position $(n-1)$.
- **Difference of Consecutive Sums Shortcut:** To quickly extract any individual term a_n from a sum formula S_n:

$$a_n = S_n - S_{n-1} \quad (\text{Valid for } n \geq 2)$$

You

Coding-decoding: letter and number substitution patterns

Gemini 3.6 Thinking

D. E. Shaw Logical Reasoning Cheat Sheet

Coding-Decoding: Letter & Number Substitution Patterns

Coding-decoding questions in advanced assessment tests evaluate pattern recognition under extreme time constraints. Cracking these problems in under 20 seconds requires replacing manual alphabet counting with **Position Anchors**, **Symmetry Maps**, and **Matrix Inspection Protocols**.

---

1. Alphabetic Reference Maps & Fast Anchors

Forward & Reverse Positional Value Map

Letter	A	B	C	D	E	F	G	H	I	J	K	L	M
Forward (Pos)	1	2	3	4	5	6	7	8	9	10	11	12	13
Reverse (Rev)	26	25	24	23	22	21	20	19	18	17	16	15	14

Letter	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Forward (Pos)	14	15	16	17	18	19	20	21	22	23	24	25	26

Letter	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Reverse (Rev)	13	12	11	10	9	8	7	6	5	4	3	2	1

Golden Anchor Formulas

- **Reverse Position Transformation:**

$$\text{Reverse Position} = 27 - \text{Forward Position}$$

- **Opposite Pair Identity:** Two letters are exact opposites if their positional sum equals 27:

$$\text{Pos}(L_1) + \text{Pos}(L_2) = 27$$

Essential Speed Benchmarks

- **EJOTY (Multiples of 5):** $\text{E}(5) - \text{J}(10) - \text{O}(15) - \text{T}(20) - \text{Y}(25)$
- **CFLORUX (Multiples of 3):** $\text{C}(3) - \text{F}(6) - \text{L}(9) - \text{O}(12) - \text{R}(15) - \text{U}(18) - \text{X}(21)$

Opposite Pair Mnemonic Cheat Sheet

\mathbf{A-Z} \text{ (AZure)} & \mathbf{B-Y} \text{ (BYe)} & \mathbf{C-X} \text{ (CruX)} & \mathbf{D-W} \text{ (Dew)} \\ \mathbf{E-V} \text{ (EVening)} & \mathbf{F-U} \text{ (FUn)} & \mathbf{G-T} \text{ (GT road)} & \mathbf{H-S} \text{ (High School)} \\ \mathbf{I-R} \text{ (Indian Railway)} & \mathbf{J-Q} \text{ (Jack Queen)} & \mathbf{K-P} \text{ (KanPur)} & \mathbf{L-O} \text{ (LOve)} \\ & & \mathbf{M-N} \text{ (MaN)} & \end{array}$$ --- ## 2. Letter-to-Letter Substitution Blueprints ### Pattern 1: Shift & Cyclic Rotations (Forward / Backward) **Mechanism:** Each letter is shifted by a constant $+k$, a progressive value $(+1, +2, +3, \dots)$, or alternating values $(+k, -k)$. * **Cyclic Wrap-Around Rule:** $$\text{New Pos} = [(\text{Original Pos} - 1 \pm k) \pmod{26}] + 1$$ > **Example:** $Z (+3) \to C$; $A (-2) \to Y$. --- ### Pattern 2: Block Reversal & Structural Swaps **Mechanism:** The word is divided into 2 or 3 equal blocks, and positions are swapped inside each block before applying shifts. #### 1. Odd-Length Word Center Pivot: * Word of length 5 ($L_1 L_2 L_3 L_4 L_5$): * Swap halves: $(L_2 L_1) \mathbf{L_3} (L_5 L_4)$ #### 2. Cross-Swap Pair Shift: ``` Original: M A T H \ / \ / \/ \/ Coded: B N I U (Swapped adjacent pairs and added +1) ``` --- ### Pattern 3: Vowel-Consonant Bifurcation **Mechanism:** Rules apply differently depending on whether the letter is a vowel or a consonant. * **Vowels (A, E, I, O, U):** Shifted by $+2$ or replaced by next sequential vowel ($A \to E \to I \to O \to U$). * **Consonants:** Shifted by -1 or replaced by opposite 27-pair. --- ## 3. Letter-to-Number Coding Blueprints When letters are transformed into numbers, test the numerical transformations in this exact priority sequence: ``` [LETTER-TO-NUMBER CODING DECODER] | +--------------------------+--------------------------+ | | Single Value Output? Multi-Digit String Output? (e.g., WORD = 52) (e.g., WORD = 2315184) | | +--> 1. Direct Sum: Σ Pos(Li) +--> 1. Concatenated Positions +--> 2. Reverse Sum: Σ (27 - Pos(Li)) (Pos(L1) | Pos(L2) | ...) +--> 3. Sum × Word Length +--> 2. Concatenated Digital Roots +--> 4. (Vowels Sum) - (Consonants Sum) (e.g., K(11) -> 2) ``` --- ### Master Formula Catalog for Number Outputs #### 1. Direct Positional Sum $$\text{Code} = \sum_{i=1}^{N} \text{Pos}(L_i)$$ #### 2. Reverse Positional Sum $$\text{Code} = \sum_{i=1}^{N} (27 - \text{Pos}(L_i)) = 27N - \sum_{i=1}^{N} \text{Pos}(L_i)$$ #### 3. Length Multiplier Modulation $$\text{Code} = \left(\sum_{i=1}^{N} \text{Pos}(L_i) \right) \times N$$ #### 4. Vowel/Consonant Weighted Sum $$\text{Code} = (\text{Sum of Vowel Positions} \times k_1) + (\text{Sum of Consonant Positions} \times k_2)$$ #### 5. Digital Root / Single Digit Compression Replace each letter's positional value with its repeated sum of digits until a single digit remains: $$\text{Digital Root}(n) = 1 + (n - 1) \pmod 9$$ * $\text{Pos}(K) = 11 \to 1 + 1 = \mathbf{2}$ * $\text{Pos}(T) = 20 \to 2 + 0 = \mathbf{2}$ * $\text{Pos}(R) = 18 \to 1 + 8 = \mathbf{9}$ --- ## 4. Advanced & Hybrid Variations ### Variation 1: Direct Fictional Language / Symbol Elimination **Scenario:** Given multiple statements with common words and codes: * *"krane blip viz"* = *"blue clear sky"* * *"viz zop torm"* = *"clear night rain"* #### Elimination Blueprint: 1. Isolate repeated words across sentences: **"clear"** appears in both. 2. Isolate repeated code tokens across lists: **"viz"** appears in both. 3. Infer directly: $\mathbf{\text{"clear"} = \text{"viz"}}$. 4. Remove identified pairs to solve remaining tokens. --- ### Variation 2: Matrix / Grid Position Coordinates **Scenario:** Letters are displayed in a 5×5 matrix. Letters are coded by **[Row Index][Column Index]**. $$\begin{array}{c|ccccc} & 0 & 1 & 2 & 3 & 4 \\ \hline 0 & \text{A} & \text{B} & \text{C} & \text{D} & \text{E} \\ 1 & \text{F} & \text{G} & \text{H} & \text{I} & \text{J} \\ 2 & \text{K} & \text{L} & \text{M} & \text{N} & \text{O} \\ \end{array}$$ * $\text{H} = 12$ (Row 1, Col 2) * $\text{N} = 23$ (Row 2, Col 3) > **Shortcut:** When solving grid matrix questions with multiple choice options, **always

test options from right to left (last digit of the word first)**. Incorrect options are usually filtered out faster from the end. --- ### Variation 3: Conditional Rule-Based Coding **Scenario:** A string of numbers/letters is converted according to specific conditional precedence rules: * *Rule 1:* If the first element is an odd number and the last is a vowel, swap their codes. * *Rule 2:* If two even numbers appear consecutively, code both as `*`. * *Rule 3:* If a vowel is immediately preceded and followed by a consonant, code the vowel as `#`. #### Blueprint: 1. Do **NOT** write down the full standard code first. 2. Scan conditions in order (Rule 1 $\to$ Rule 2 $\to$ Rule 3) and flag affected positions. 3. Apply overrides **only** to the flagged positions; fill remaining positions with standard values. --- ## 5. 15-Second Pattern Identification Protocol ``` Step 1: Compare lengths of input word and output code. └──► If Equal: Letter shift, block swap, or 1-to-1 position coding. └──► If Single Number: Positional sum, reverse sum, or length multiplier. └──► If Double Length: Full 2-digit position concatenation (e.g., A=01, B=02). Step 2: Check the First and Last Letters of the input/output. └──► If First maps to Last: Reversal pattern (Word reversed + shift). └──► If First maps to Opposite: 27-Pair complementary pattern. Step 3: Check Vowel Positions. └──► If vowels are unaltered or follow a distinct step size (+1 for vowels, +2 for consonants), the pattern is Vowel-Consonant split. ``` --- ## 6. Common Traps & Precision Checklist ``` [TRAP ALERT 1: Forward vs. Backward Alphabetic Overlap] • Watch out for cyclic shifts near edges (A, B and Y, Z). • Moving backward from B by 3 positions: B(2) -> A(1) -> Z(26) -> Y(25). • Formula verification: 2 - 3 = -1 => 26 + (-1) = 25 (Y). ``` ``` [TRAP ALERT 2: "Opposite Pair" Sum Fallacy] • The sum of a letter and its opposite is ALWAYS 27, NOT 26. • A(1) + Z(26) = 27 • M(13) + N(14) = 27 ``` * **Letter Count Multiplier Trap:** If $\text{CAT} = 72$ and $\text{DOG} = 78$: * $\text{CAT} = (3 + 1 + 20) \times 3 = 24 \times 3 = 72$ * $\text{DOG} = (4 + 15 + 7) \times 3 = 26 \times 3 = 78$ * The multiplier 3 is the **length of the word**, not a fixed constant. Test a longer word (e.g., BIRD, length 4) with $N=4$, not 3.

You

Series completion: number and letter series with non-obvious rules

Gemini 3.6 Thinking

D. E. Shaw Logical Reasoning Cheat Sheet

Advanced Series Completion: Non-Obvious Number & Letter Patterns

In high-speed assessments, standard arithmetic and geometric progressions are rare. Advanced tests like D. E. Shaw's feature **higher-order polynomial differences, recursive digital operations, interleaved hybrid logic, and prime/factorial gap sequences**.

---

1. Advanced Number Series Blueprints

Pattern 1: Method of Finite Differences (Polynomial Series)

If a series grows moderately but without a constant multiplier, construct a **Difference Tree** ($\Delta^1, \Delta^2, \Delta^3$).

```
Series:      4      11      25      48      82     129
  Δ¹:          +7      +14     +23     +34     +47
  Δ²:             +7      +9      +11     +13      (AP Gap)
  Δ³:                +2      +2      +2           (Constant)
```

Fundamental Theorem of Finite Differences:

- If the **k-th difference (Δ^k) is constant**, the general term T_n is a polynomial of degree k:

$$T_n = an^k + bn^{k-1} + \cdots + z$$

- Δ^1 constant $\implies$ Linear $(an + b)$

- Δ^2 constant $\implies$ Quadratic ($an^2 + bn + c$)
- Δ^3 constant $\implies$ Cubic ($an^3 + bn^2 + cn + d$)

Pattern 2: Multiplicative-Additive Hybrid Recurrences

Each term is generated by multiplying the previous term by a factor and adding/subtracting a modifier:

$$T_n = (k \cdot T_{n-1}) \pm f(n)$$

Operation Type	Recurrence Formula	Example Sequence Structure
Constant Multiplier, Variable Addend	$T_n = 2T_{n-1} + n$	$3, 8, 19, 42, 89 \ldots (\times 2 + 2, \times 2 + 3, \times 2 + 4)$
Variable Multiplier, Constant Addend	$T_n = n \cdot T_{n-1} - 1$	$2, 3, 7, 27, 134 \ldots (\times 2 - 1, \times 3 - 1, \times 4 - 1)$
Alternating Operation Engine	$T_n = (T_{n-1} \times a)$ then $(T_{n-1} + b)$	$5, 15, 18, 54, 57 \ldots (\times 3, +3, \times 3, +3)$

Pattern 3: Square & Cube Offsets ($n^k \pm f(n)$)

Instead of pure squares or cubes, terms are shifted by indices, primes, or previous terms:

- **Index-Shifted Powers:** $T_n = n^2 \pm n \implies n(n \pm 1)$
- **Cubed Index Shifts:** $T_n = n^3 \pm n^2$
- **Prime Power Offsets:** $T_n = n^2 + p_n$ (where p_n is the n-th prime)

> **Quick Identifiers:**
>
> - $2, 6, 12, 20, 30, 42 \cdots \implies n^2 + n$ or $n(n+1)$
> - $0, 6, 24, 60, 120, 210 \cdots \implies n^3 - n$ or $(n-1)n(n+1)$
> - $2, 12, 36, 80, 150, 252 \cdots \implies n^3 + n^2$ or $n^2(n+1)$

Pattern 4: Factorial & Prime Gaps

- **Factorial Offsets:** $T_n = n! \pm k$ or $T_n = n! \pm n$
  - *Sequence:* $0, 1, 5, 23, 119 \implies (n! - 1)$
- **Prime Gap Multipliers:** Gaps between terms follow the prime number sequence ($2, 3, 5, 7, 11, 13 \ldots$).
- **Composite-Only Gaps:** Gaps skip all prime numbers and use only composite numbers ($4, 6, 8, 9, 10, 12 \ldots$).

---

2. Non-Standard & Digit-Level Series Mechanics

Pattern 5: Multi-Term Memory (Fibonacci & Tribonacci Variants)

Terms depend on the sum or product of two or more preceding terms:

$$T_n = f(T_{n-1}, T_{n-2}, \ldots)$$

- **Product Fibonacci:** $T_n = T_{n-1} \times T_{n-2}$
  - *Sequence:* $2, 3, 6, 18, 108, 1944 \ldots$
- **Square-Sum Recurrence:** $T_n = T_{n-1}^2 - T_{n-2}^2$

- **Tribonacci (3-Term Memory):** $T_n = T_{n-1} + T_{n-2} + T_{n-3}$
  - *Sequence:* $1, 1, 2, 4, 7, 13, 24, 44 \ldots$

Pattern 6: Intra-Digit Operations (Self-Referential Series)

The next term depends on arithmetic operations performed on the **digits of the current term**:

- **Sum of Digits Addend:** $T_n = T_{n-1} + \text{SumOfDigits}(T_{n-1})$
  - *Example:* $14 \to 14 + (1 + 4) = 19 \to 19 + (1 + 9) = 29 \to 29 + (2 + 9) = 40$
- **Product of Digits Addend:** $T_n = T_{n-1} + \text{ProductOfDigits}(T_{n-1})$
  - *Example:* $23 \to 23 + (2 \times 3) = 29 \to 29 + (2 \times 9) = 47$
- **Digital Root Cycles (Modulo 9):** $T_n = \text{DigitalRoot}(T_{n-1}^2)$

Pattern 7: Interleaved (Twin/Triple) Series

Two or three independent series are combined into a single sequence:

```
Combined:   A1,  B1,  A2,  B2,  A3,  B3 ...
Position:   1    2    3    4    5    6

Odd Positions (A):   2,   4,   8,  16 ... (Powers of 2)
Even Positions (B):  81, 27,   9,   3 ... (Divided by 3)
```

> **Dead Giveaway:** If a series oscillates up and down repeatedly ($12, 5, 15, 8, 18, 11 \ldots$), split it into odd and even positions immediately.

---

3. Advanced Letter & Hybrid Series Mechanics

Pattern 1: Non-Linear Positional Step Sizes

Letter steps follow non-linear mathematical sequences (Primes, Fibonacci, Squares):

- **Prime-Step Letter Progression:**
  - Positions: $A(1) \xrightarrow{+2} C(3) \xrightarrow{+3} F(6) \xrightarrow{+5} K(11) \xrightarrow{+7} R(18) \xrightarrow{+11} C(29 \equiv 3)$
- **Fibonacci-Step Letter Progression:**
  - Positions: $+1, +1, +2, +3, +5, +8 \ldots$

Pattern 2: Complementary 27-Pair & Symmetric Loops

- **Alternating Opposite Pairs:** Letters alternate between forward position and the reverse 27-pair:
  - $A(1) \to Z(26) \to C(3) \to X(24) \to E(5) \to V(22)$
- **Center Pivot Mirroring:**
  - First half of alphabet maps to second half in reverse ($A \to Z, B \to Y, \ldots$).

Pattern 3: Alpha-Numeric Value Operations

Letter positions are manipulated using mathematical functions to yield numbers, or vice versa:

$$\text{Value} = \frac{\text{Pos}(L_1) \times \text{Pos}(L_2)}{k}$$

- *Example:* $\text{C} - 9 - \text{F} - 36 - \text{I} - 81 \ldots$ (Letter position squared: $3^2 = 9, 6^2 = 36, 9^2 = 81$)

- *Example:* $B - D - 8, C - E - 15, D - F - 24$ (Product of letter positions: $2 \times 4 = 8, 3 \times 5 = 15, 4 \times 6 = 24$)

Pattern 4: Semantic / Non-Alphabetic Letter Series

Letters do not represent positions in the alphabet, but the **first letters of a known natural sequence**:

- **Calendar/Time:** $J, F, M, A, M, J, J, A, S, O, N, D$ (Months)
- **Ordinals:** $F, S, T, F, F, S, S, E, N, T$ (First, Second, Third, Fourth...)
- **Planets / Constants:** M, V, E, M, J, S, U, N (Mercury, Venus, Earth...)
- **Prime Words:** T, T, F, S, E, T, T, N (Two, Three, Five, Seven, Eleven...)

---

4. 15-Second Pattern Isolation Protocol

```
                    [ EVALUATING A NUMBER SERIES ]
                                  |
        +-------------------------+-------------------------+
        |                                                   |
Does it move MONOTONICALLY?                    Does it OSCILLATE up & down?
(Strictly increases or decreases)                           |
        |                                                   +--> Split into INTERLEAVED
        v                                                        Series (Odd/Even positions).
Is growth RAPID (Exponential-like)?
        |
        +---> YES: Test Multiplicative-Additive (T_n = a*T_{n-1} ± b),
        |          Factorials (n! ± k), or Product Recurrences.
        |
        +---> NO:  Calculate Difference Tree (Δ¹, Δ², Δ³).
                   └──► Constant Δ¹ = AP
                   └──► Constant Δ² = Quadratic (n² ± k)
                   └──► Constant Δ³ = Cubic (n³ ± k)
                   └──► If Δ values are Primes/Squares = Sub-layer pattern.
```

---

5. Common Traps & Edge Cases

```
[ TRAP ALERT 1: Digital Root Modulo Wrap ]
• When working with positional letters near Z (e.g., +7 from V(22)):
  22 + 7 = 29 => 29 - 26 = 3 (C).
• Formula: Pos = [(Current + Step - 1) mod 26] + 1
```

```
[ TRAP ALERT 2: The "Prime 1" Trap ]
• 1 is NOT a prime number.
• Prime sequence ALWAYS starts at 2: (2, 3, 5, 7, 11, 13, 17...).
• Composite sequence starts at 4: (4, 6, 8, 9, 10, 12, 14...).
```

- **Visual Alignment Trap:** Do not assume a series uses standard alphabetic positions if step differences look completely erratic ($+3, -5, +12, -2$). Immediately test for **Semantic Sequences** (First letters of numbers/months) or **Complementary 27-Pairs**.

You

Seating arrangements: linear and circular, with multiple constraints simultaneously

D. E. Shaw Logical Reasoning Cheat Sheet

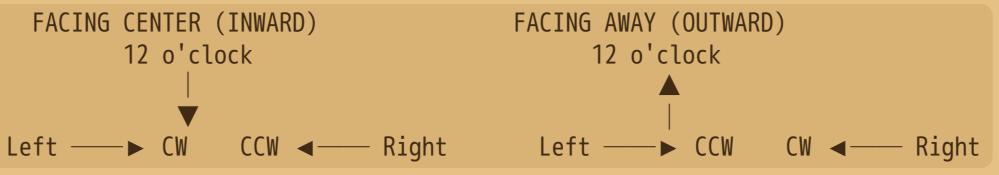
Seating Arrangements: Multi-Constraint Linear & Circular Mechanics

Advanced seating arrangement puzzles in assessment tests involve multi-layered constraints: mixed facing directions (Inward/Outward, North/South), dual parallel lines, multi-variable attributes (Professions/Colors), and family tree integration.

---

1. Directional Mechanics & Spatial Orientation

1. Circular Geometry Mechanics



Facing Direction	Movement Direction	Directional Output
Center / Inward	**Clockwise (CW)**	Moves **LEFT**
Center / Inward	**Counter-Clockwise (CCW)**	Moves **RIGHT**
Away / Outward	**Clockwise (CW)**	Moves **RIGHT**
Away / Outward	**Counter-Clockwise (CCW)**	Moves **LEFT**

---

2. Directly Opposite Positions (The Even-Seat Rule)

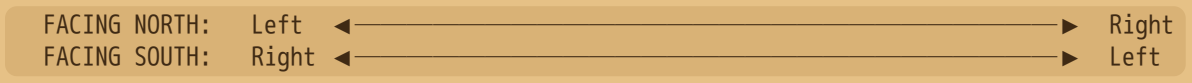
- **Even Seats ($N = 6, 8, 10$):** Every seat has a uniquely defined **directly opposite seat**.

$$\text{Opposite Seat Distance} = \frac{N}{2} \text{ seats away (in either direction)}$$

- *Example ($N = 8$):* Person at Position 1 is directly opposite Position $1 + \frac{8}{2} = 5$.
- **Odd Seats ($N = 5, 7, 9$): NO ONE** sits directly opposite anyone. Do not attempt to draw opposite lines for odd-numbered arrangements.

---

3. Linear Orientation Mechanics



- **Same Facing Direction:** If A and B face the same direction, A's left is B's left.
- **Opposite Facing Direction:** If A and B face opposite directions, A's left is B's right.

---

2. Standardized Notation System for Scratch Work

To eliminate re-reading dense question text, translate all clues immediately into standard visual shorthand:

Textual Clue Statement	Shorthand Notation	Visual Meaning
A sits 3rd to the right of B	$B \xrightarrow{+3} A$ or $B _ _ A$	2 people between B and A

Textual Clue Statement	Shorthand Notation	Visual Meaning
A sits immediate left of B	(AB) or $\underline{A\,B}$	Tied block; order is fixed
A and B are immediate neighbors	$(A \leftrightarrow B)$	Tied block; order unknown (AB or BA)
A does not sit adjacent to B	$A \neq B$ or $A \not\leftrightarrow B$	Cannot occupy adjacent slots
A sits at an extreme end	$A = \text{End}$	Must occupy index 1 or N
A faces North / Inward	$A \uparrow$ or $A(\text{In})$	Facing direction fixed
A faces South / Outward	$A \downarrow$ or $A(\text{Out})$	Facing direction fixed
A and B face opposite directions	$A \neq_{dir} B$	One is North/In, other South/Out

---

3. Linear Arrangement Blueprints

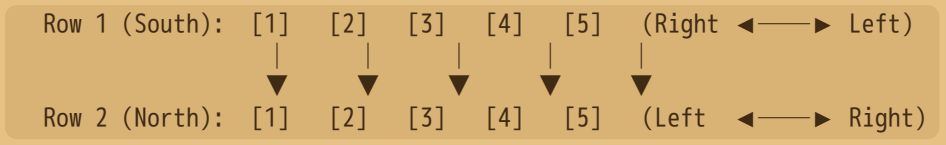
Blueprint 1: Mixed Facing Linear Rows (North / South)

Scenario: N people sit in a row; some face North, others face South.

1. **Fixed Anchor Search:** Locate clues giving absolute end positions (e.g., *"A sits 4th from the left end"*).
2. **Directional Deduction Rule:**
   - If X faces North and Y is k-th to the right of X, place Y at $\text{Pos}(X) + k$.
   - If X faces South and Y is k-th to the right of X, place Y at $\text{Pos}(X) - k$.
3. **Boundary Test:** If placing a person exceeds the row boundaries (< 1 or $> N$), discard that orientation case instantly.

Blueprint 2: Dual Parallel Rows

Scenario: Row 1 (facing South, looking down) and Row 2 (facing North, looking up).



- **Opposite Alignment Rule:** If A (Row 1) faces B (Row 2), they occupy identical column indices.
- **Diagonal Relations:**
  - *"A sits diagonal to B"*: A and B are at opposite extreme ends of the two rows.
  - *"A faces the person who sits 2nd to the left of B"*: First locate B in Row 2, move 2 positions left to find target T, then place A in Row 1 directly opposite T.

Blueprint 3: Unknown Number of Persons in a Line

Scenario: A variable number of people sit in a straight line facing North.

1. Do **NOT** draw fixed slots $1 \ldots N$.
2. Build **relative distance chains** using given gap clues:

$$\text{Block 1: } A _ _ _ B \quad (4 \text{ slots})$$

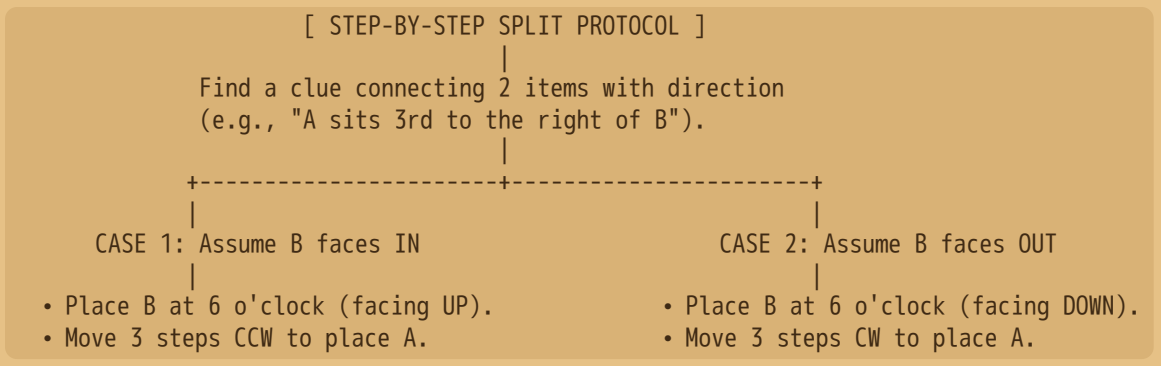
3. Overlay overlapping blocks (e.g., *"C sits 2nd to the right of A"* $\implies A _ C _ B$).
4. Only count total seats after all negative end constraints are applied.

---

4. Circular Arrangement Blueprints

Blueprint 1: Inward / Outward Mixed Facing Circle

Scenario: 8 people sit around a circular table; some face the center, others face outward.



- **Branching Limit:** Maintain a maximum of **2 parallel working diagrams**. The invalid case will violate a boundary or facing constraint within 3-4 steps.

Blueprint 2: Circular + Multi-Variable Attribute (Person + Profession + City)

Do not rely on the circular diagram alone to store attribute data. Combine the spatial diagram with a **Secondary Elimination Table**:

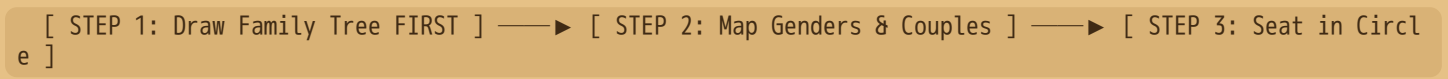
Seat Position	Person	Facing Direction	Profession / Attribute
1	A	Inward	Engineer
2	B	Outward	Doctor
3	C	Inward	Analyst

- **Linking Rule:** Use negative attribute clues (e.g., *"The Doctor does not face Inward"*) to immediately eliminate candidates from specific seat positions in the table.

Blueprint 3: Circular Arrangement Layered with Blood Relations

Scenario: Family members sit around a table with constraints based on relationships (e.g., *"No wife sits adjacent to her husband"*).

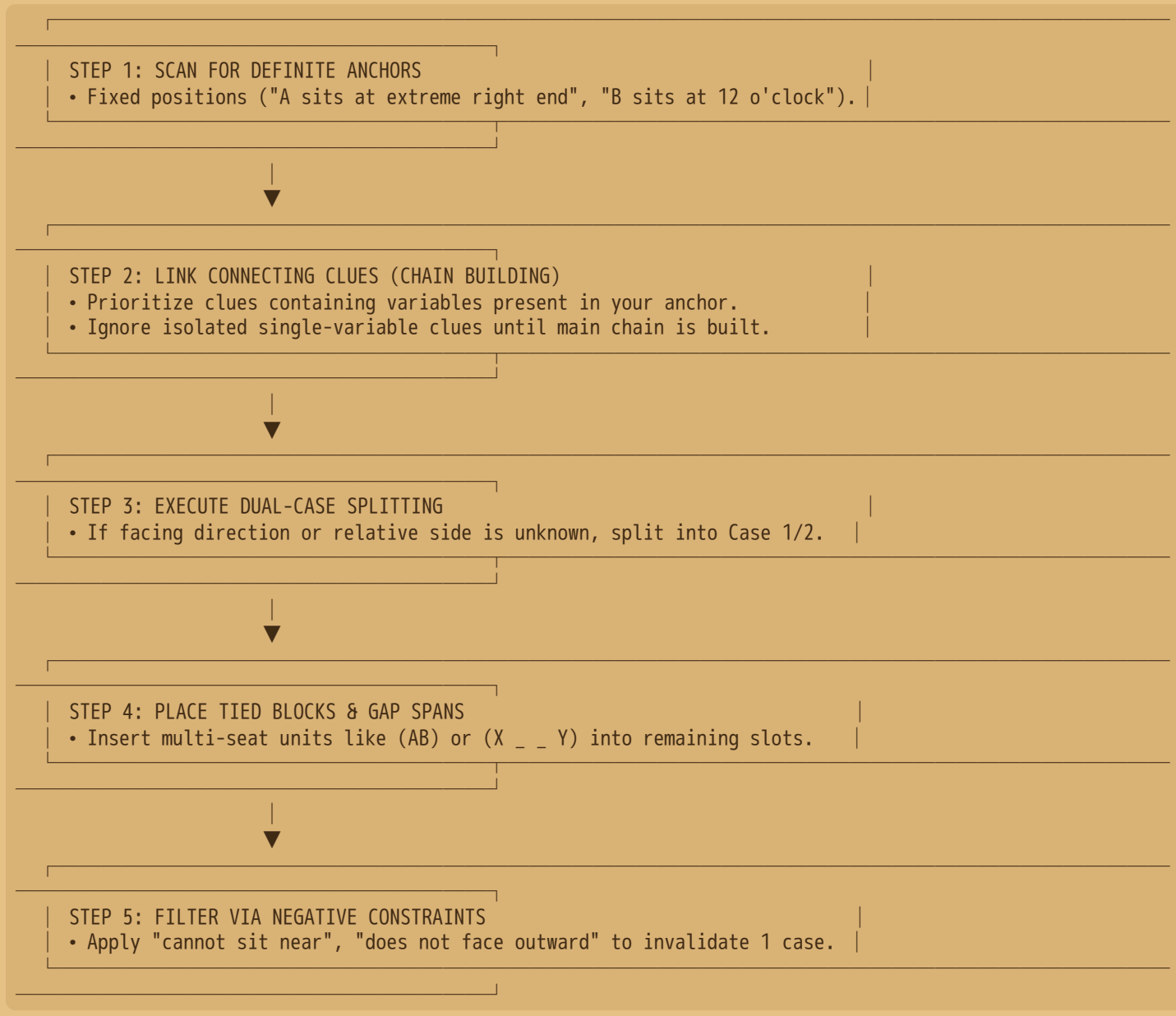
Execution Order Rule:



1. Solve the **Family Tree structure entirely** before drawing the circular seating table.
2. Determine:
   - Total number of Males (M) and Females (F).
   - Exact marital pairs: $(M_1 - F_1), (M_2 - F_2)$.
3. Apply relational seating rules:
   - *"A sits 2nd to the left of his wife"* $\implies$ Place A (M), locate seat 2 steps to the left, assign seat to A's specific wife F_A.

---

5. The 5-Step Resolution Algorithm



6. Common Traps & Edge Cases

```
[ TRAP ALERT 1: "Immediate Left" vs. "Left of" ]
• "A sits to the left of B"  ——► A can be ANYWHERE to the left of B (1, 2, or 3+ slots away).
• "A sits immediate left of B" ——► A must be EXACTLY adjacent to B (0 slots between).
```

```
[ TRAP ALERT 2: "Facing Opposite" vs. "Sitting Opposite" ]
• Sitting Opposite: Geometrically directly across the table (Requires EVEN seat count).
• Facing Opposite: Lines of sight point at each other.
  - Two people sitting geometrically opposite face each other ONLY IF BOTH FACE INWARD.
  - If one faces Inward and one faces Outward, they face the SAME direction overall.
```

```
[ TRAP ALERT 3: Gender / Relation Neighbor Constraints ]
• "No two females sit together" ——► Females must be separated by at least 1 male (F M F M F).
• In an 8-person circle with 4 males and 4 females, "No two females sit together"
  FORCES an exact alternating pattern: F - M - F - M - F - M - F - M.
```

You

Blood relations: multi-generation family trees

D. E. Shaw Logical Reasoning Cheat Sheet

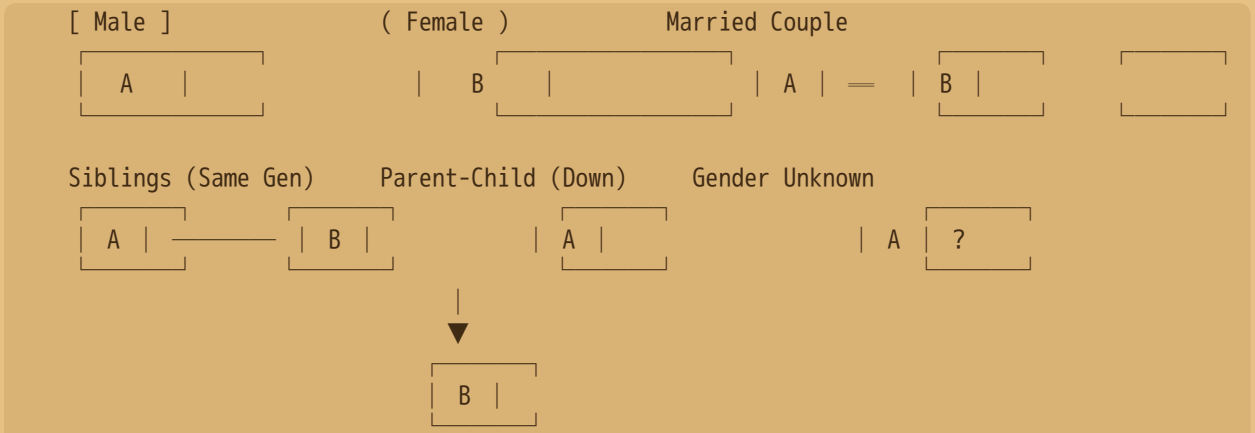
Blood Relations: Multi-Generation Family Trees & Coded Logic

Blood relation puzzles in high-speed assessment tests measure your ability to parse complex relational structures without getting lost in ambiguous phrasing. Master the **Standard Symbol Notation**, **Generation Gap Indexing**, and **Coded Option Elimination** to crack these problems rapidly.

---

1. Standard Visual Shorthand & Symbol Notation

Do not write out relationship names on your scratch paper. Use a unified diagrammatic notation to build family trees instantly:



Relationship Dynamic	Visual Diagram Shorthand	Gender / Structural Rule
Male	$[A]$ or $A(+)$	Enclose in a box or tag with $(+)$.
Female	(A) or $A(-)$	Enclose in a circle or tag with $(-)$.
Married Couple	$A \iff B$ or $A = B$	Double line; connects different genders (unless unspecified).
Siblings	$A - B$	Single horizontal line at the same vertical level.
Parent-Child	A $\vert$ B	Vertical line (A is 1 generation above B).

---

2. Generation Gap Indexing System

Assign a numerical **Generation Index (G)** relative to the reference person ($G = 0$) to compute generation gaps algebraically:

Relational Tier	Generation Index (G)	Examples
Grandparents Tier	$+2$	Grandfather, Grandmother, Grand-uncle, Grand-aunt
Parents Tier	$+1$	Father, Mother, Uncle, Aunt, Father-in-law, Mother-in-law
Self / Peer Tier	0	Brother, Sister, Cousin, Spouse, Brother-in-law, Sister-in-law
Children Tier	-1	Son, Daughter, Nephew, Niece, Son-in-law, Daughter-in-law

Relational Tier	Generation Index (G)	Examples
Grandchildren Tier	-2	Grandson, Granddaughter

---

3. Problem Variations & Step-by-Step Blueprints

Variation 1: Pointing / Direct Dialogue Statements

Scenario: *"Pointing to a photograph, a man says, 'She is the daughter of the only son of my grandfather.' How is the woman related to the man?"*

Blueprint: Backwards Unwinding Protocol

1. Locate the possessive anchor word: **"my"** (belongs to the speaker $\rightarrow$ **Man**).
2. Split the statement into reverse sequential clauses moving outward from **"my"**:

$$\text{Man} \xrightarrow{\text{my grandfather}} G_1 \xrightarrow{\text{only son of } G_1} G_2 \xrightarrow{\text{daughter of } G_2} \text{Target (Woman)}$$

3. Evaluate step-by-step:
   - *Step 1:* "my grandfather" $\rightarrow$ Speaker's Grandfather ($G = +2$).
   - *Step 2:* "only son of my grandfather" $\rightarrow$ Speaker's Father ($G = +1$).
   - *Step 3:* "daughter of [my father]" $\rightarrow$ Speaker's Sister ($G = 0$).
4. **Result:** The woman is the **Sister** of the man.

Variation 2: Coded Blood Relations ($A + B$, $A \times B$)

Scenario:

- $A + B \implies A$ is the father of B ($G = +1$, Male)
- $A - B \implies A$ is the sister of B ($G = 0$, Female)
- $A \times B \implies A$ is the mother of B ($G = +1$, Female)
- $A \div B \implies A$ is the brother of B ($G = 0$, Male)

"Which expression indicates that P is the Maternal Aunt of Q?"

Blueprint: Two-Pass Rapid Filter (Gender + Generation Gap)

- **Filter 1: Target Gender Check**
  - Target: P is a **Maternal Aunt** $\implies$ P MUST be **Female**.
  - Scan options for operator immediately following P:
    - If $P + X \implies P$ is male $\rightarrow$ **Eliminate**.
    - If $P \div X \implies P$ is male $\rightarrow$ **Eliminate**.
    - If P is at the very end of the expression (e.g., $X + Y \div P$), P's gender is usually **unknown** $\rightarrow$ **Eliminate** (unless married couple context forces gender).
- **Filter 2: Net Generation Gap (ΔG) Check**
  - Target: Maternal Aunt is 1 generation above $Q \implies \mathbf{\Delta G = +1}$.
  - Calculate net sum of generation values from P to Q for remaining options:

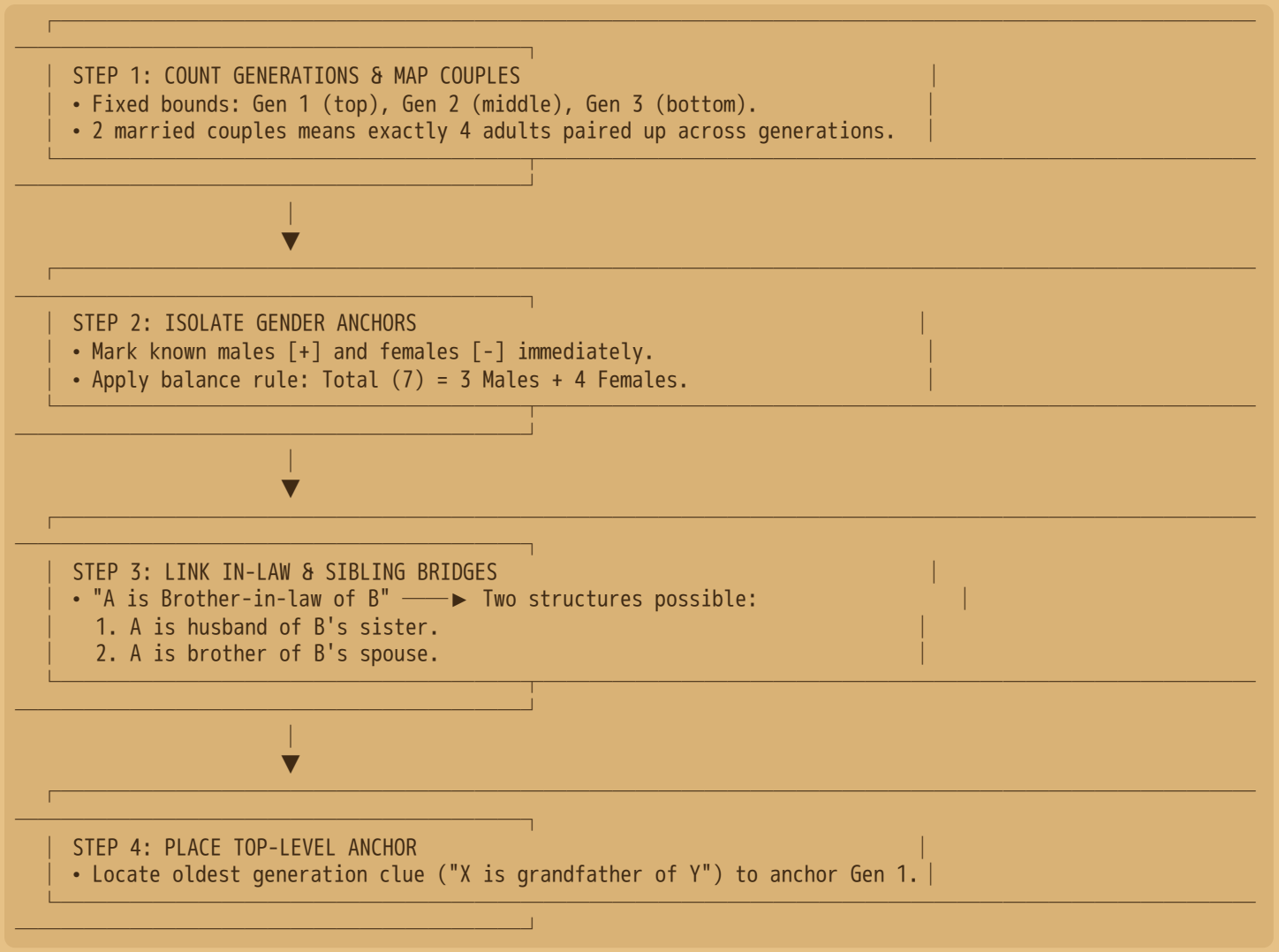
$$\text{Option A: } P - R \times S + Q \implies (0) + (+1) + (+1) = +2 \quad (\text{Grandmother tier} \times)$$

Option B: $P - R \times Q \implies (0) + (+1) = +1 \quad (\text{Aunt tier} \checkmark)$

Variation 3: Complex Multi-Generation Tree Puzzles

Scenario: A family of 7 people (A through G) across 3 generations contains 2 married couples, 3 males, 4 females, and specific profession/attribute clues.

Blueprint: Structural Constraint Matrix



---

4. Mathematical Gender & Generation Algebra

For any coded path $P \xrightarrow{e_1} X_1 \xrightarrow{e_2} X_2 \cdots \xrightarrow{e_n} Q$:

$\Delta G_{\text{total}} = \sum_{i=1}^{n} \Delta G(e_i)$

Operator Relationship	Gender of First Element	Generation Shift (ΔG)
Father / Mother	Male / Female	$+1$
Uncle / Aunt	Male / Female	$+1$
Brother / Sister / Spouse	Male / Female	0
Son / Daughter	Male / Female	-1
Nephew / Niece	Male / Female	-1

---

5. Essential Terminology & Ambiguity Resolution

Term	Precise Definition	Potential Trap / Misconception
"Only Son"	A person has **only 1 son**, but may have multiple daughters.	Does NOT mean "only child".
"Only Child"	A person has **exactly 1 child** (no brothers or sisters).	Means 0 brothers and 0 sisters.
"Paternal"	Related through the **Father's** side of the family.	Paternal Uncle = Father's Brother.
"Maternal"	Related through the **Mother's** side of the family.	Maternal Aunt = Mother's Sister.
Brother-in-law	Husband of one's sister **OR** Brother of one's spouse.	Requires checking both structural branches.
Sister-in-law	Wife of one's brother **OR** Sister of one's spouse.	Requires checking both structural branches.
Sibling	Brother or Sister (sharing same parents).	Does NOT include cousins.

---

6. Common Traps & Precision Checklist

```
[ TRAP ALERT 1: Gender Assumption by Name ]
• NEVER assume gender based on a person's name (e.g., "Rahul" is not automatically Male).
• Determine gender ONLY via explicit relational descriptors ("father", "sister", "wife").
```

```
[ TRAP ALERT 2: Tail Element Gender Ambiguity in Coded Logic ]
• In an expression like A + B - C × D, D is the LAST element.
• Unless D is defined as someone's husband/wife, D's gender CANNOT be determined.
• Do not pick an option where the question asks for "How is D related to A?"
  if D's gender remains unstated—the answer is likely "Cannot be determined".
```

```
[ TRAP ALERT 3: Spurious In-Law Connections ]
• "A is the Father-in-law of B" ——► A is male, and A's child is married to B.
• Ensure you draw the intermediate spouse node; do not link A to B directly.
```

You

Analytical puzzles: multi-variable constraint satisfaction problems

Gemini 3.6 Thinking

D. E. Shaw Logical Reasoning Cheat Sheet

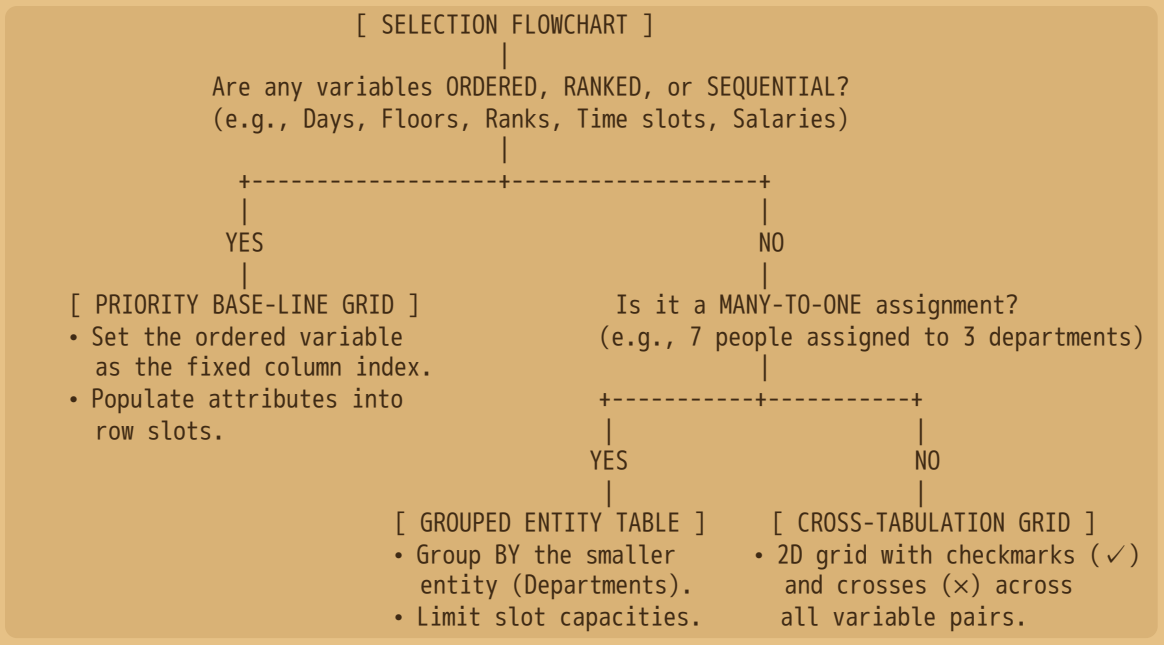
Analytical Puzzles: Multi-Variable Constraint Satisfaction Problems (CSPs)

Complex multi-variable analytical puzzles evaluate structural organization under strict time limits. Cracking these problems requires transforming unstructured conditions into **Matrix Grids**, **Domain-Reduction Engines**, and **Constraint Transfer Graphs**.

---

1. Grid Selection Blueprint & Data Representation

The first step in any analytical puzzle is choosing the correct matrix architecture based on variable types:



---

2. Standard Matrix Architectures

1. Priority Base-Line Grid (Ordered / Sequential Problems)

Used when one variable has a natural sequence (Days of the week, Floors 1–8, Years).

Floor / Day	Person	Department	City / Preference	Item
Floor 4	A	Finance	New York	Laptop
Floor 3	B	IT	London	Tablet
Floor 2	C	Marketing	Tokyo	Phone
Floor 1	D	HR	Paris	Monitor

---

2. Grouped Entity Table (Capacity-Constrained Assignment)

Used when N items are distributed across M categories ($N > M$, e.g., 8 employees across 3 departments: HR, Tech, Sales), where each category has minimum/maximum capacity limits.

Department	Assigned Entities (N)	Gender / Attributes	Variable Constraints
HR (Min 2, Max 3)	$A(+),\ E(-)$	1 Male, 1 Female	No member likes Red
Tech (Min 2, Max 3)	$B(+),\ D(+),\ G(-)$	2 Males, 1 Female	Includes highest earner
Sales (Min 2, Max 3)	$C(-),\ F(+)$	1 Female, 1 Male	Must have Tokyo traveler

---

3. Cross-Tabulation Elimination Matrix (Attribute Matching)

Used when matching multiple independent sets of attributes (e.g., 4 People, 4 Pets, 4 Car Models) without numerical sequence or grouping.

& \text{Dog} & \text{Cat} & \text{Bird} & \text{Fish} & \text{Sedan} & \text{SUV} & \text{EV} & \text{Truck} \\ \hline \text{Alice} & \mathbf{\checkmark} & \times & \times & \times & \times & \mathbf{\checkmark} & \times & \times \\ \text{Bob} & \times &

\times & \mathbf{\checkmark} & \times & \times & \mathbf{\checkmark} & \times \\ \text{Charlie} & \times & \mathbf{\checkmark} & \times & \times & \mathbf{\checkmark} & \times & \times & \times \\ \text{David} & \times & \times & \times & \mathbf{\checkmark} & \times & \times & \times & \mathbf{\checkmark} \\ \end{array}$$ * **Row/Column Elimination Rule:** When a cell receives a $\checkmark$, instantly place $\times$ in all other cells along its row and column for that variable block. --- ## 3. Constraint Mechanics & Visual Shorthand To avoid reading the problem description repeatedly, convert all raw textual rules into visual shorthand: | Rule Type | Textual Example | Visual Shorthand | Algorithmic Meaning | | :--- | :--- | :--- | | **Exact Tie** | "A works in the same department as B" | $(A = B)$ | Bound entity block; move together. | | **Negative Tie** | "A does not belong to Marketing" | $A \neq \text{Mkt}$ | Domain reduction; eliminate slot option. | | **Adjacency / Gap** | "A lives 2 floors above B" | $\begin{matrix} A \\ _ \\ B \end{matrix}$ | Relative offset: $\text{Pos}(A) = \text{Pos}(B) + 2$. | | **Conditional** | "If A goes to Paris, then B takes the Train" | $A_{\text{Paris}} \implies B_{\text{Train}}$ | Contrapositive holds: $B_{\neg \text{Train}} \implies A_{\neg \text{Paris}}$. | | **Group Bounds** | "At least 2 and at most 3 people in HR" | $2 \le \|\text{HR}\| \le 3$ | Track capacity counters during placement. | | **Attribute Split** | "No two females work in Sales" | $\|\text{Sales}_{\text{Female}}\| \le 1$ | Capacity restriction on gender attribute. | --- ## 4. The 5-Step Domain-Reduction Solver Algorithm ```

┌──────────────

Common Traps & Precision Checklist ``` [TRAP ALERT 1: Contrapositive Inversion Fallacy] • Statement: "If A goes to Boston, then B goes to Chicago" (A_Bos ⇒ B_Chi). • WRONG INFERENCE: "If A DOES NOT go to Boston, then B DOES NOT go to Chicago" (A_¬Bos ⇏ B_¬Chi). • CORRECT CONTRAPOSITIVE: "If B DOES NOT go to Chicago, then A DOES NOT go to Boston" (B_¬Chi ⇒ A_¬Bos). ``` ``` [TRAP ALERT 2: "Above / Below" vs. "Immediately Above / Below"] • "A lives above B" ——► Pos(A) > Pos(B) (Difference can be 1, 2, 3... floors). • "A lives immediately above B" ——► Pos(A) = Pos(B) + 1 (Difference is EXACTLY 1). ``` * **Capacity Boundary Exhaustion:** When allocating entities, keep a running counter of assigned males/females or category totals. If a group requires "at least 2 females" and 2 females are already placed in other groups, the remaining slots in that group **must** be filled by males.

You

Syllogisms: Venn diagram logic and deductive inference

Gemini 3.6 Thinking

D. E. Shaw Logical Reasoning Cheat Sheet

Syllogisms: Categorical Logic, Venn Diagrams, & Deductive Inference

Solving syllogisms with high speed and zero error requires bypassing intuitive guessing. Master the **100–50 Value Distribution Rules**, the **Complementary Pair Engine**, and **Modern Quantifier Translations** ("Only a few", "Only") to evaluate complex deductive arguments rapidly.

---

1. Categorical Propositions & Value Mapping

Every standard syllogistic statement connects a **Subject** (S) and a **Predicate** (P). In competitive assessments, propositions are evaluated using either **Venn Diagrams** or the **100–50 Distribution Method**.

Core Proposition Classifications

Code	Statement Type	Formal Representation	Venn Overlap	Subject (S) Value	Predicate (P) Value
A	Universal Affirmative	*"All S are P"*	$S \subseteq P$	100 (Distributed)	50 (Undistributed)
E	Universal Negative	*"No S is P"*	$S \cap P = \emptyset$	100 (Distributed)	100 (Distributed)
I	Particular Affirmative	*"Some S are P"*	$S \cap P \neq \emptyset$	50 (Undistributed)	50 (Undistributed)
O	Particular Negative	*"Some S are not P"*	$S \setminus P \neq \emptyset$	50 (Undistributed)	100 (Distributed)

The 100–50 Analytical Method Rules

Think of premise values as **Income** and conclusion values as **Expense**:

$$\text{Valid Deduction Rule: Value in Conclusion} \leq \text{Value in Premise}$$

- A term valued at **50 in the premises** CANNOT become **100 in the conclusion**.
- A term valued at **100 in the premises** can be **50 or 100 in the conclusion**.

---

2. Master Syllogistic Rules Engine (Direct Mediate Inference)

When combining two premises to form a conclusion via a **Middle Term** (M):

```
  Premise 1:  S ——— M
  Premise 2:  M ——— P
  Conclusion: S ——— P
```

The 6 Fundamental Deduction Rules

1. **Middle Term Distribution Rule:** The Middle Term (M) **must be distributed (value = 100) at least once** across the two premises. If $M_{\text{Premise 1}} = 50 \text{ AND } M_{\text{Premise 2}} = 50 \implies \text{No Valid Mediate Conclusion}$
2. **Premise Polarity Rules:**
   - $\text{Positive} + \text{Positive} \implies \text{Positive Conclusion}$
   - $\text{Positive} + \text{Negative} \implies \text{Negative Conclusion}$
   - $\text{Negative} + \text{Negative} \implies \text{No Valid Conclusion}$
3. **Particular Premise Rules:**
   - $\text{Universal} + \text{Universal} \implies \text{Universal or Particular Conclusion}$
   - $\text{Universal} + \text{Particular} \implies \text{Particular Conclusion}$
   - $\text{Particular} + \text{Particular} \implies \text{No Valid Conclusion}$ (i.e., $I + I, I + O, O + O$ yield no standard conclusion).
4. **Term Retention Rule:** If a term is distributed (100) in the conclusion, it **must** be distributed (100) in its respective premise.

---

3. Modern Quantifiers & Phrase Translations

Modern reasoning tests frequently use non-standard quantifiers. Translate them instantly into standard categorical types before drawing diagrams or evaluating values:

```
                [ MODERN QUANTIFIER TRANSLATOR ]
                                 |
    +----------------------------+----------------------------+
    |                                                         |
"ONLY A IS B"                                   "ONLY A FEW A ARE B"
    |                                                         |
    v                                                         v
All B are A                                     Dual Condition:
(B is EXCLUSIVELY tied to A;                    1. Some A are B   (I-type: Positive)
 no other category can overlap B).              2. Some A are not B (O-type: Negative)
```

Textual Phrase	Standard Categorical Translation	Type	Notes / Exceptions
"Only A is B"	**All B are A**	**A**	B cannot overlap with any third variable C.
"Only a few A are B"	**Some A are B** AND **Some A are not B**	**I + O**	Generates BOTH positive and negative boundaries.
"Few / Seldom / Barely A are B"	**Some A are not B**	**O**	Negative connotation $\implies$ Particular Negative.

Textual Phrase	Standard Categorical Translation	Type	Notes / Exceptions
"At least / Almost / Most A are B"	**Some A are B**	**I**	Standard Particular Affirmative.
"All A can never be B"	**No A is B**	**E**	Definite universal negative statement.
"Each / Every / Any A is B"	**All A are B**	**A**	Standard Universal Affirmative.

---

4. Possibility Cases & "Either-Or" Complementary Pairs

1. Possibility Rules Engine

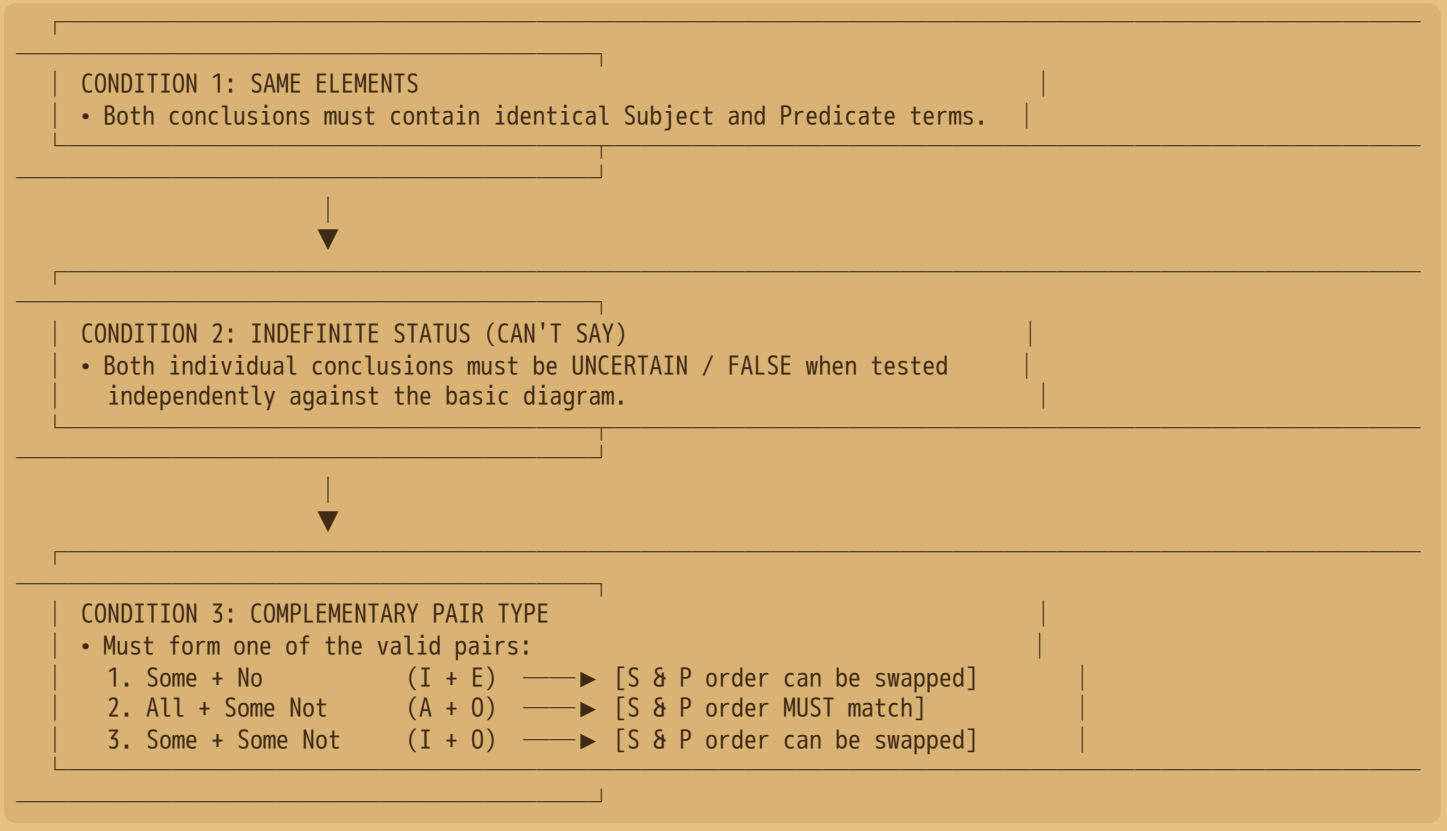
A conclusion stating a **"Possibility"** (e.g., *"All A being B is a possibility"*) must be evaluated against the **Basic (Minimal Overlap)** diagram and **Possible** diagrams:

- If the definite relationship is **definitely TRUE** in all cases $\implies$ Possibility is **FALSE** (it is already a certainty).
- If the definite relationship is **definitely FALSE** (violates a given statement) $\implies$ Possibility is **FALSE**.
- If the definite relationship is **UNKNOWN / UNCERTAIN** ("Can't Say") $\implies$ Possibility is **TRUE**.

---

2. "Either-Or" Complementary Pair Protocol

For an **"Either Conclusion I or Conclusion II follows"** relationship to hold, ALL THREE of the following conditions must be satisfied simultaneously:



> **Crucial Exception:** The pair **"All + No"** ($A + E$) **is NOT a complementary pair** for an "Either-Or" choice, because both statements can be false simultaneously (Contraries).

---

5. Visual Venn Diagram Verification Blueprint

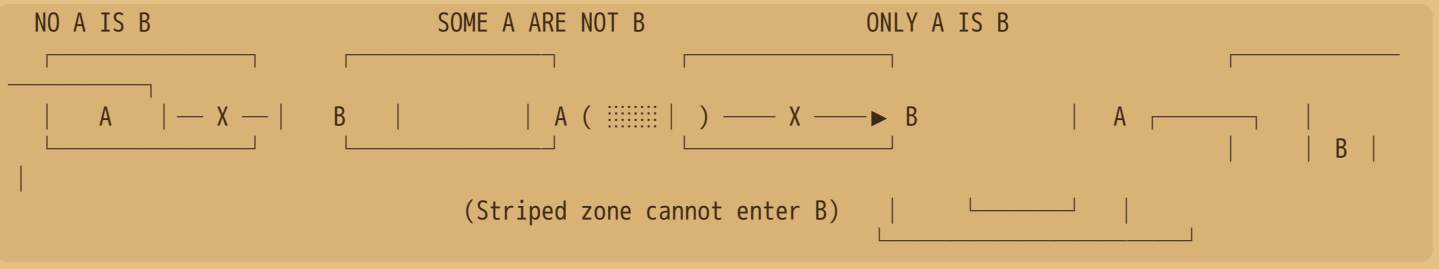
When solving multi-statement syllogisms with 3 or 4 variables, follow this systematic visual protocol:

```
Step 1: Draw the BASIC DIAGRAM (Minimal Overlap)
        • Represent "All A are B" as A inside B.
        • Represent "Some A are B" with minimum necessary overlap.
        • Represent "No A is B" with an explicit cross-bar line (A ——— X ——— B).

Step 2: Evaluate Definite Conclusions
        • A definite conclusion is VALID if and only if it holds TRUE
          in the Basic Diagram AND all possible alternate diagrams.

Step 3: Evaluate Possibility Conclusions
        • A possibility conclusion is VALID if you can construct at least ONE
          valid alternate diagram where the condition holds without violating
          the original premises.
```

Visual Representation of Exclusions



---

6. Common Traps & Precision Checklist

```
[ TRAP ALERT 1: The "Only A is B" Multi-Variable Interference ]
• Given: "Only A is B" and "Some A are C".
• WRONG INFERENCE: Assuming "Some B can be C" is a valid possibility.
• CORRECT TRUTH: B is LOCKED inside A. B CANNOT overlap with C under ANY
  circumstance. "Some B being C is a possibility" is FALSE.
```

```
[ TRAP ALERT 2: Reversing "Some A are not B" ]
• Statement: "Some A are not B".
• WRONG INFERENCE: Assuming "Some B are not A" automatically follows.
• Deductive logic DOES NOT support conversion of O-type statements.
```

```
[ TRAP ALERT 3: Possibility with Absolute Knowledge ]
• Statement: "No A is B".
• Question: "Is 'Some A being B' a possibility?"
• Answer: FALSE. A possibility cannot contradict a known universal negative premise.
```

- **Subject-Predicate Swap in "All + Some Not" ($A + O$):** For an "Either-Or" case with A and O statements, terms must line up strictly:
  - *Valid:* "All X are Y" + "Some X are not Y".
  - *Invalid:* "All X are Y" + "Some Y are not X" (Does **not** form a valid complementary pair).

You

Reading comprehension: short passages with inference and vocabulary questions

Gemini 3.6 Thinking

D. E. Shaw Verbal Reasoning Cheat Sheet

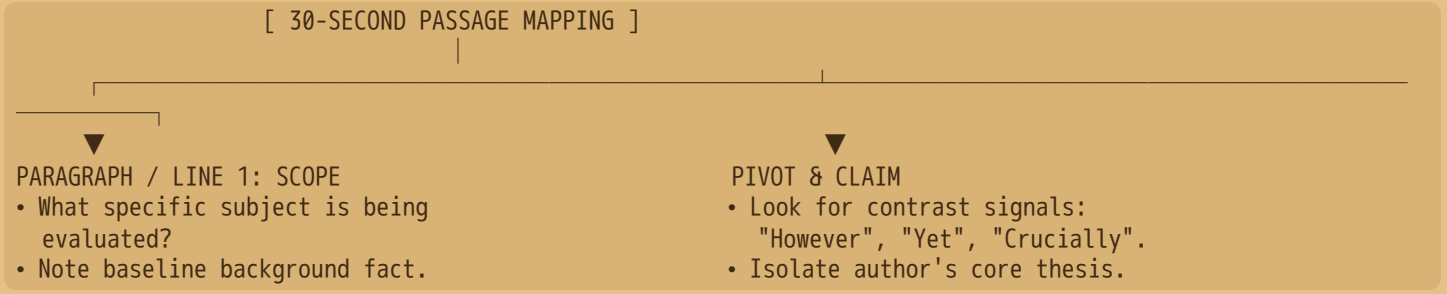
Reading Comprehension: Short Passages, Inferences, & Contextual Vocabulary

High-speed Reading Comprehension (RC) in quantitative and analytical assessments tests your ability to map structure, extract strictly bounded logical deductions, and decode context-dependent vocabulary under severe time constraints. Master the **Structural Mapping Protocol**, the **Strict Inference Filter**, and the **Blank Replacement Technique** to evaluate passages in under 60 seconds.

---

1. Passage Deconstruction & Active Reading Protocol

Do not read short passages for detailed retention; read for **Structural Architecture** and **Author Stance**.



Structural Signal Words Map

Signal words indicate structural shifts and tell you where answer choices are hidden:

Signal Category	Trigger Words	Structural Function	Action Required
Contrast / Pivot	*However, Nonetheless, Yet, Conversely, On the other hand, Granted*	Reverses direction; marks main point or counter-argument.	**Slow down.** The line following a pivot usually contains the main idea or inference anchor.
Causation	*Thus, Consequently, Hence, Therefore, Driven by, Yielding*	Indicates cause-and-effect relationship.	Check for **Reversal Traps** in options (confusing cause with effect).
Concession	*Although, Despite, In spite of, While it is true that*	Downplays one point to emphasize another.	Identify the main clause—that is what the author actually cares about.
Amplification	*Furthermore, Moreover, Indeed, In addition, Significantly*	Adds supporting evidence to existing claim.	Speed read; details here are usually distractor material.

---

2. Question-Type Mechanics & Execution Blueprints

Type 1: The Strict Inference Blueprint

An inference in standardized testing is **NOT** a guess, speculation, or reasonable extrapolation. It is an **unstated premise or logical consequence that MUST BE 100% TRUE** given the passage text.

$$\text{Passage Statement } (P) \implies \text{Valid Inference } (Q)$$

```
┌──────────────────────────────────────────────────────────────────┐
│ THE INFERENCE STRICTNESS TEST                                    │
│ • Ask yourself: "Can I imagine a world where the passage is 100% true,  │
│   but this option is FALSE?"                                     │
│                                                                  │
│   └──► YES  ──► Option is INVALID (It is a "Could Be True" distractor). │
│   └──► NO   ──► Option is VALID (It MUST BE TRUE).               │
└──────────────────────────────────────────────────────────────────┘
```

Blueprint:

1. Isolate the specific line numbers or keywords referenced in the stem.
2. Combine the passage premise with formal logical constraints (e.g., If *All A are B* and *B are C*, then *Some A are C*).
3. **Reject** options that introduce outside facts, future predictions, or moral judgements.

Type 2: Contextual Vocabulary Blueprint (Vocabulary-in-Context)

Questions asking for the meaning of a word in line X test **contextual usage**, not primary dictionary definitions. Test setters deliberate pick common words used in secondary or tertiary senses.

The "Blank Replacement Protocol":

1. Cover the target word in the passage with your finger.
2. Replace it with a placeholder: `[BLANK]`.
3. Re-read 1 sentence before and 1 sentence after to establish the **Tone (+ / - / neutral)** and **Logical Relationship**.
4. Predict your own simple replacement word for `[BLANK]`.
5. Match your prediction against the choice set.

> **Example:** *"The director attempted to **qualify** her previous statement regarding corporate earnings."*
>
> - *Step 1:* "The director attempted to `[BLANK]` her previous statement..."
> - *Step 2:* Context shows she added limitations to prevent misunderstanding.
> - *Prediction:* "limit" or "modify".
> - *Correct Choice:* **Limit / Restrain** (NOT "certify" or "authorize").

Type 3: Primary Purpose & Main Idea Blueprint

Primary Purpose options always consist of an **Action Verb + Target Scope**.

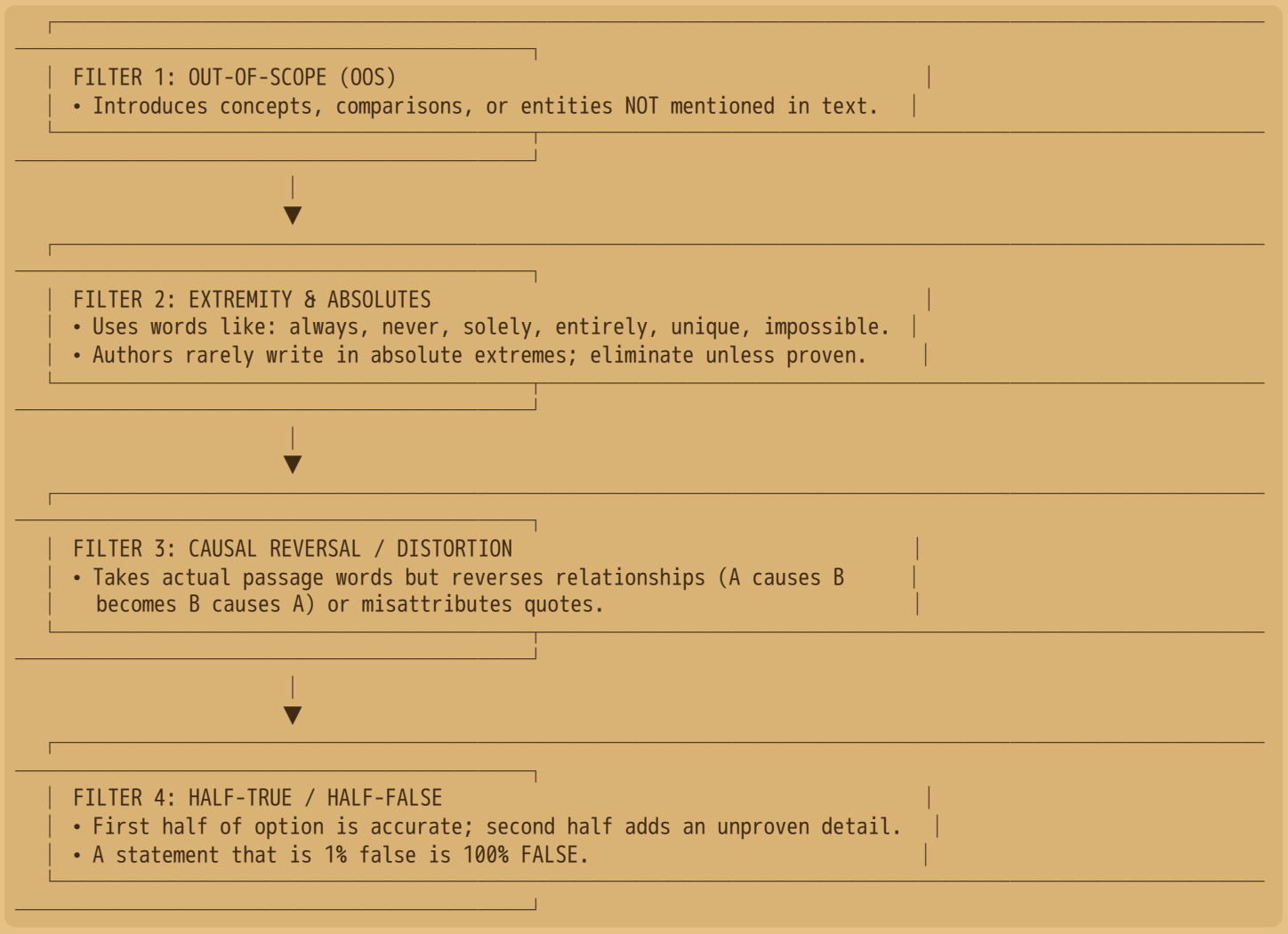
```
              [ PRIMARY PURPOSE MATCHING MATRIX ]
                              │
      ┌───────────────────────┴──────────────────────────┐
      ▼                                                  ▼
ACTION VERB MUST MATCH STANCE                  SCOPE MUST MATCH BOUNDS
• Evaluate: Neutral analysis of two sides.     • Must cover entire passage,
• Advocate: Takes a clear positive stance.       not just Paragraph 1 or
• Refute: Attacks an existing theory.            a single data point.
• Delineate: Outlines a system or process.
```

- Eliminate options where the **Verb is too aggressive** (e.g., "prove" when the author only "suggests").
- Eliminate options where the **Scope is too narrow** (focuses on a single example/detail).

---

3. The 4-Filter Option Elimination Engine

When deciding between the final two answer options, run them through these four strict filters:



---

4. Contextual Vocabulary Substitution Matrix

Save time by memorizing secondary/tertiary definitions of high-frequency aptitude test words:

Target Word	Primary / Common Meaning	Secondary Contextual Meaning (Passage Usage)	Example Context
Qualify	To meet requirements	**To limit, modify, or restrict a claim**	*"He qualified his support with conditions."*
Compromise	To reach an agreement	**To weaken, endanger, or expose to risk**	*"Security was compromised by the breach."*
Check	To inspect or verify	**To hold back, restrain, or stem growth**	*"The policy checked rapid inflation."*
Flag	A banner / symbol	**To decline in energy, vigor, or strength**	*"Interest in the project flagged."*
Plastic	Synthetic polymer material	**Adaptable, flexible, or easily shaped**	*"Neural pathways remain plastic."*
Arrest	To take into legal custody	**To stop, delay, or interrupt a process**	*"The treatment arrested disease growth."*
Base	The bottom support	**Lacking moral worth, ignoble, or crude**	*"Motivated by base financial gain."*

Target Word	Primary / Common Meaning	Secondary Contextual Meaning (Passage Usage)	Example Context
Tolerate	To endure / put up with	**To allow for variance or margin of error**	*"The engine tolerates heat swings."*

---

5. Common Traps & Precision Checklist

```
[ TRAP ALERT 1: Outside Knowledge Leakage ]
• NEVER use real-world facts to answer an RC question.
• If the passage states "The Earth is flat", then for that test, the Earth is flat.
• Any option containing true real-world facts NOT supported by text is OUT-OF-SCOPE.
```

```
[ TRAP ALERT 2: Right Answer to the WRONG Question ]
• The option contains a statement that is 100% true according to the passage,
  but it answers line 15 when the question explicitly asked about line 42.
• Always re-read the stem to confirm alignment before selecting.
```

```
[ TRAP ALERT 3: Comparative & Superlative Offsets ]
• Passage: "Company A increased its profits this quarter."
• Wrong Option: "Company A had the HIGHEST profits" or "Company A was MORE profitable than B".
• An increase (relative shift) does NOT imply superlative rank or comparison to others.
```

- **The Neutral Tone Rule:** In scientific or business passages, author tone is overwhelmingly **measured, objective, or qualified**. Options with emotional language (*indignant, ecstatic, disdainful, malicious*) are virtually always wrong.

You

Sentence correction: grammatical and usage errors

Gemini 3.6 Thinking

D. E. Shaw Verbal Reasoning Cheat Sheet

Sentence Correction: Grammatical & Usage Errors

In high-speed verbal assessments, sentence correction questions test your ability to spot structural flaws, modifier errors, and idiomatic missteps in seconds. Master the **15-Second Error Isolation Protocol**, **Parallel Structure Mechanics**, and **Modifier Alignment Rules** to solve questions without relying on how the sentence "sounds."

---

1. Subject-Verb Agreement (SVA) & Intervening Distractors

The subject and verb must agree in number (singular vs. plural), regardless of intervening phrases, inverted word orders, or compound structures.

1. The Prepositional & Modifier Blind Spot

Ignore all prepositions, relative clauses, and appositives separating the core subject from the verb:

$$\text{Subject} + \cancel{[\text{Prepositional Phrase / Relative Clause}]} + \text{Verb}$$

- **Incorrect:** *"The collection of rare manuscripts, ancient coins, and historical relics **were** auctioned off."*

- **Correct:** *"The collection ~~of rare manuscripts...~~ **was** auctioned off."*

---

2. Complex & Indefinite Pronoun Rules

Pronoun Category	Tokens	Rule	Example
Always Singular	*Each, Every, Everyone, Someone, Anyone, Neither, Either, Nobody*	Takes a **singular verb**, even if followed by a plural prepositional object.	*"Each of the candidates **is** qualified."*
Always Plural	*Both, Few, Many, Several*	Takes a **plural verb**.	*"Several of the reports **were** missing."*
SANAM Group (Variable)	***S**ome, **A**ny, **N**one, **A**ll, **M**ore / **M**ost*	Verb depends on the **object of the preposition** (O).	$\text{SANAM} + \text{of } O_{\text{sing}} \implies \text{Verb}_{\text{sing}}$ $\text{SANAM} + \text{of } O_{\text{plur}} \implies \text{Verb}_{\text{plur}}$

> **SANAM Examples:**
>
> - *"All of the **water was** spilled."* (Water = Singular)
> - *"All of the **books were** destroyed."* (Books = Plural)

---

3. Compound Subjects & Conjunctions

- A and B: Takes a **plural** verb (*"The manager and the analyst **are** reviewing the file"*), UNLESS both terms refer to the same person/thing (*"My supervisor and mentor **is** retiring"*).
- A or B / Either A or B / Neither A nor B: The verb agrees with the **closest subject** (B):

$$\text{Neither } A_{\text{singular}} \text{ nor } B_{\text{plural}} + \text{Verb}_{\text{plural}}$$

$$\text{Neither } A_{\text{plural}} \text{ nor } B_{\text{singular}} + \text{Verb}_{\text{singular}}$$

---

2. Modifier Placement & Alignment Rules

Modifiers must sit as close as possible to the exact noun/pronoun they modify.

1. The Dangling Modifier (Introductory Participle Trap)

When a sentence begins with a modifying phrase (participial or prepositional), the noun immediately following the comma **must be the actor doing that action**.

```
[ INTRODUCTORY MODIFIER PHRASE ] ,  [ NOUN / SUBJECT ]  +  [ VERB ]
                                  |          ▲
                                  └──────────┘
                          (Must be the DIRECT ACTOR of the phrase)
```

- **Incorrect:** *"Walking through the financial district, the tall skyscrapers impressed the investors." (Dangling: Were the skyscrapers walking? No.)*
- **Correct:** *"Walking through the financial district, the **investors** were impressed by the tall skyscrapers."*

---

2. Misplaced Limiting Modifiers

Words like *only, almost, nearly, merely, barely* modify the term **immediately following them**:

- *"She **only** eats vegetables."* $\implies$ Eating is the *only* action she does to vegetables (she doesn't cook or sell them).
- *"She eats **only** vegetables."* $\implies$ Vegetables are the *only* thing she eats.

---

3. Relative Pronoun Modifiers (*Which* vs. *That* vs. *Who*)

- **That (Restrictive / Essential):** Essential to sentence meaning; **no comma** preceding it.
  - *"The portfolio **that contains tech equities** outperformed the market."*
- **Which (Non-restrictive / Non-essential):** Adds extra context; **requires a comma**.
  - *"The main portfolio*, which contains tech equities,** outperformed the market."*
- **Who / Whom:** Must refer to **people** (never use *which* or *that* for humans).

---

3. Parallelism & Correlative Conjunctions

Items in a list, comparison, or split structure must maintain identical grammatical forms (Noun with Noun, Verb with Verb, Gerund with Gerund).

Structural Symmetry Matrix

$$\text{Correct Parallel Form: } X \text{ AND } Y \implies [\text{Grammatical Structure of } X] \equiv [\text{Grammatical Structure of } Y]$$

```
INCORRECT:  The executive enjoys  [ golfing ] ,  [ to ski ] ,  and  [ dinner out ] .
                                    Gerund         Infinitive         Noun Phrase

CORRECT:    The executive enjoys  [ golfing ] ,  [ skiing ] ,  and  [ dining out ] .
                                    Gerund         Gerund             Gerund
```

Correlative Conjunction Split Alignment

For paired conjunctions (*Not only... but also*, *Either... or*, *Both... and*, *Neither... nor*, *Rather than*), the structural element immediately following the first marker must mirror the structural element immediately following the second marker:

- **Incorrect:** *"The firm **not only** values integrity **but also** hard work." (After "not only": verb "values"; after "but also": noun "hard work")*
- **Correct:** *"The firm values **not only** [integrity] **but also** [hard work]." (After both markers: Nouns)*

---

4. Tense Consistency, Sequences, & Subjunctive Mood

1. Past Perfect Sequence (The "Earlier Past")

When describing two events that occurred in the past, use the **Past Perfect ($had + V_3$)** for the earlier event, and **Simple Past (V_2)** for the later event:

$$\text{Timeline: } [\text{Earlier Event: } \mathbf{Had + V_3}] \longrightarrow [\text{Later Event: } \mathbf{Simple\ Past\ (V_2)}] \longrightarrow [\text{Present}]$$

- **Correct:** *"By the time the exchange **opened** (V_2), the market **had already dropped** ($had + V_3$) 200 points."*

---

2. The Subjunctive Mood (Hypotheticals & Demands)

- **Hypothetical / Contrary-to-Fact Conditionals:** Use **"were"** (never *was*) regardless of subject singular/plural status:
  - *"If the analyst **were** (not was) present, she would explain the model."*
- **Command / Request Verbs (*Demand, Recommend, Insist, Request, Suggest*):** Requires **"that" + Subject + Base Verb (V_1)** without *should*:

- **Incorrect:** *"The auditor demanded that he **is** present / **should be** present."*
- **Correct:** *"The auditor demanded that he **be** present."*

---

5. Pronoun Case & Agreement Engine

1. Subjective vs. Objective Pronoun Case

Case Tier	Subjective (Acts)	Objective (Receives Action / Object of Prep)
Singular	*I, He, She, It, You, Who*	*Me, Him, Her, It, You, Whom*
Plural	*We, They, Who*	*Us, Them, Whom*

The "Who vs. Whom" He/Him Substitution Rule:

- Replace the target word with **He/They** $\implies$ Use **WHO**.
- Replace the target word with **Him/Them** $\implies$ Use **WHOM**.

> **Example:** *"The engineer [who / whom] wrote the algorithm won the award."*
>
> - *Test:* **He** wrote the algorithm (Valid) $\implies$ Use **WHO**.
>
> **Example:** *"The analyst [who / whom] the director promoted was praised."*
>
> - *Test:* The director promoted **him** (Valid) $\implies$ Use **WHOM**.

---

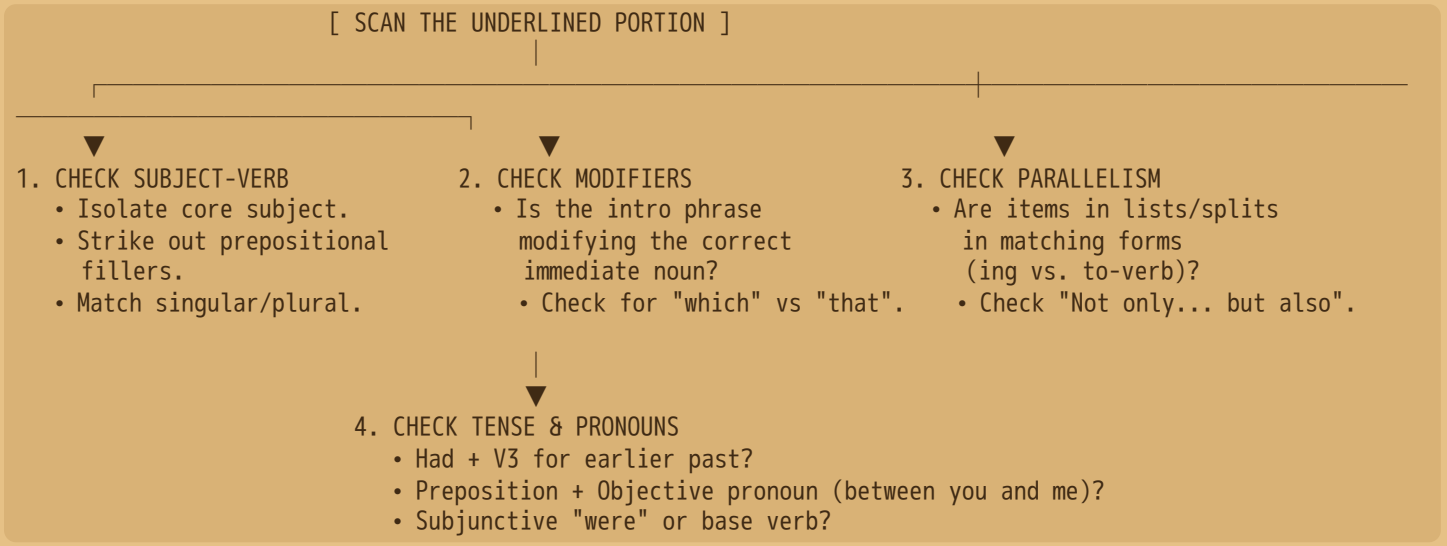
2. Idiomatic Prepositional Pronoun Case

Prepositions (*between, among, with, for, to*) MUST take **objective case pronouns**:

- **Incorrect:** *"Between you and **I**, the strategy will fail."*
- **Correct:** *"Between you and **me** ($Prep + Obj$), the strategy will fail."*

---

6. The 15-Second Sentence Correction Protocol



---

7. High-Frequency Idiomatic & Usage Traps

```
[ TRAP ALERT 1: "Like" vs. "As / As If" ]
• LIKE is a preposition ——► Followed ONLY by Nouns / Noun Phrases (No verbs).
  - Example: "He runs LIKE a deer." (Deer = Noun)
• AS / AS IF are conjunctions ——► Followed by full Clauses (Subject + Verb).
  - Incorrect: "He manages the fund like his predecessor did."
  - Correct: "He manages the fund AS his predecessor did." (His predecessor did = Clause)
```

```
[ TRAP ALERT 2: "Due To" vs. "Because Of" ]
• DUE TO modifies NOUNS (Equivalent to "caused by").
  - Example: "The delay was DUE TO rain." (Modifies delay)
• BECAUSE OF modifies VERBS / ACTIONS.
  - Incorrect: "The event was cancelled due to it rained."
  - Correct: "The event was cancelled BECAUSE OF the rain." (Modifies cancelled)
```

```
[ TRAP ALERT 3: Countable vs. Uncountable Quantifiers ]
• Countable (Items you can count): FEWER, NUMBER, MANY.
  - "Fewer shares", "A number of errors".
• Uncountable (Bulk mass/degree): LESS, AMOUNT, MUCH.
  - "Less liquidity", "A large amount of capital".
```

- **"Compared To" vs. "Compared With":**
  - **Compared to:** Used to highlight similarities between fundamentally *different* things (*"He compared the stock market to a roller coaster"*).
  - **Compared with:** Used to analyze differences/similarities between *similar* things (*"Compared with Q1 sales, Q2 sales rose 10%"*).

You

Synonyms and antonyms: context-sensitive usage, not pure definition recall

Gemini 3.6 Thinking

D. E. Shaw Verbal Reasoning Cheat Sheet

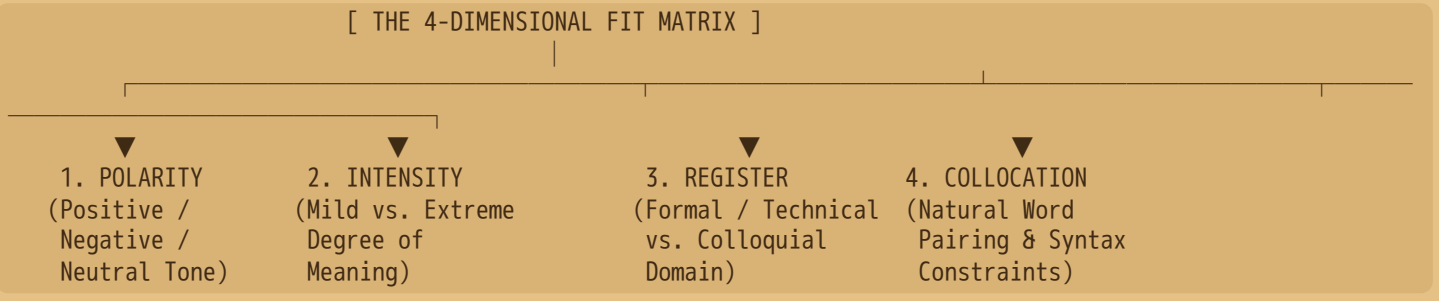
Synonyms & Antonyms: Context-Sensitive Usage & Nuance Mapping

Advanced assessment tests rarely evaluate pure dictionary definitions. Instead, they test **context-sensitive usage**, where a word's meaning is dictated by tone, domain register, intensity level, and collocation constraints. Solving these questions requires moving beyond static memorization to master **Contextual Blank-Replacement**, **Polysemous Domain Shifting**, and **Intensity Matching**.

---

1. Core Dimensions of Contextual Nuance

When evaluating synonyms and antonyms within a passage or sentence, words must match across four distinct dimensions:



1. Polarity & Tone Alignment

A word's connotation must match the emotional or evaluative orientation of the context:

- *Neutral:* **Assertive** (confident statement)
- *Negative:* **Aggressive / Imperious** (overbearing statement)
- *Positive:* **Resolute** (determined statement)

2. Intensity & Degree Spectrum

Synonyms and antonyms must align on the same intensity tier. Replacing a mild word with an extreme word creates a distortion.

$$\text{Intensity Tier Check: } \text{Intensity}(W_{\text{Context}}) \equiv \text{Intensity}(W_{\text{Option}})$$

```
[ MILD ] ──────────► [ MODERATE ] ──────────► [ EXTREME ]
Dislike              Animosity                Abhorrence
Frugal               Thrifty                  Miserly
Quiet                Reticent                 Taciturn
```

3. Domain Register & Field-Specific Semantics

Words change definitions when transferred between domain registers (Finance, Biology, Law, Everyday Speech):

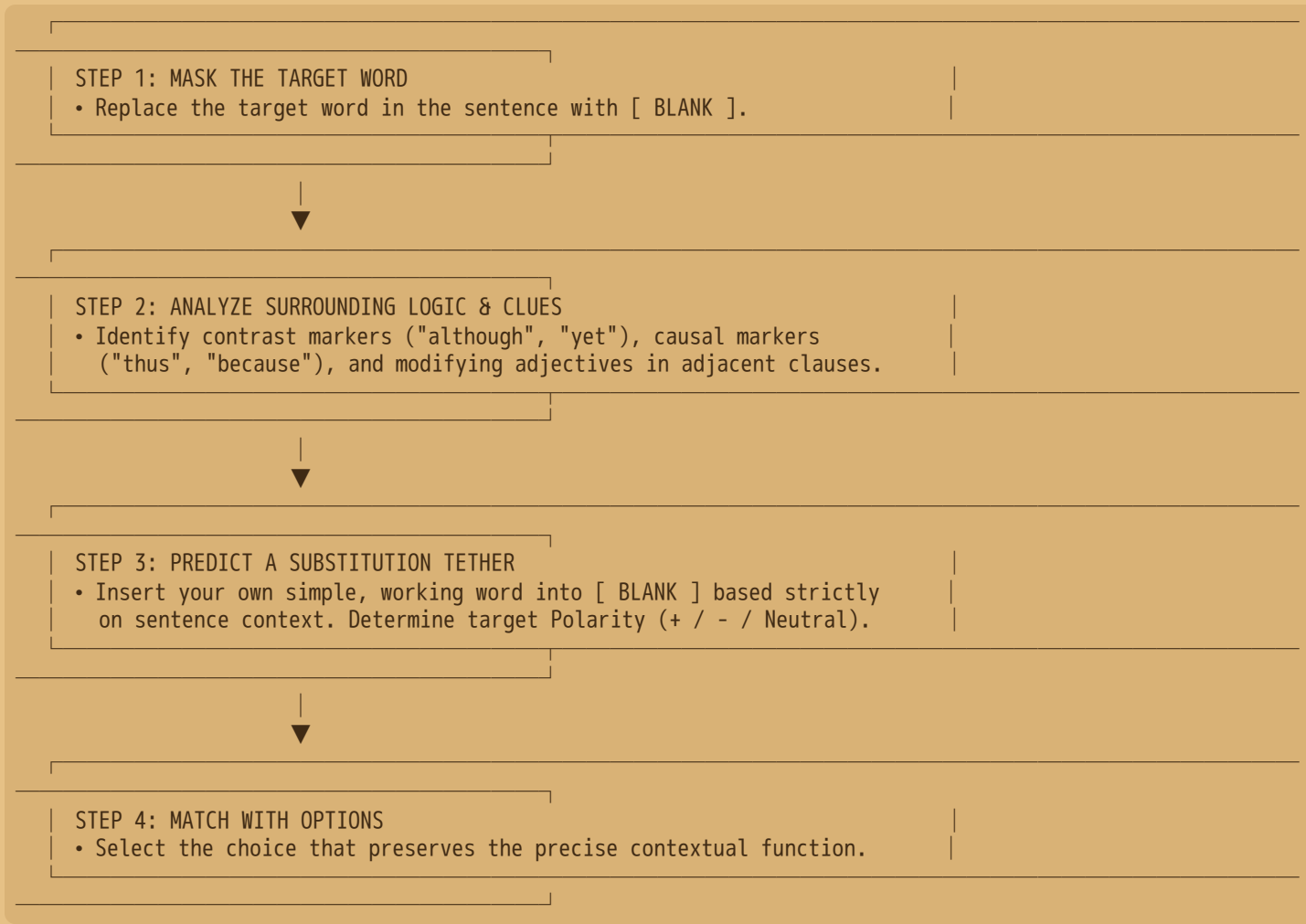
- **Discount:**
  - *Everyday Register:* A reduction in price.
  - *Epistemic/Analytical Register:* To dismiss, minimize, or ignore an idea ($\text{Synonym} = \text{Disregard}$).
- **Husband:**
  - *Everyday Register:* A male spouse.
  - *Resource Management Register:* To conserve or manage economically ($\text{Synonym} = \text{Economize}$).

---

4. Contextual Decoding Blueprints

Blueprint 1: The Contextual Blank-Replacement Algorithm

Do not evaluate the underlined target word in isolation. Strip it from the sentence to eliminate baseline definition bias:



> **Example:** *"Given the volatile nature of the energy markets, the chief risk officer urged the board to **hedge** their exposure."*
>
> 1. Mask word: "...urged the board to `[BLANK]` their exposure."
> 2. Clue analysis: "volatile nature" $\Longrightarrow$ danger/risk; "risk officer" $\Longrightarrow$ protection.
> 3. Prediction: "protect" or "safeguard".
> 4. Option match: **Safeguard / Counterbalance** (NOT "skirt" or "equivocate").

Blueprint 2: Collocation & Syntactic Compatibility Test

Two words are true contextual synonyms only if they can accept the same prepositional or syntactic complements:

Target Word	Valid Collocation / Preposition	Incorrect Substitute	Valid Contextual Synonym
Mitigate	$[\text{Target}] + \text{Direct Object (Risk/Damage)}$	Alleviate (used primarily for pain/distress)	**Attenuate / Moderate**
Abide	$[\text{Target}] + \textbf{by} + \text{Rule}$	Obey (does not take "by")	**Comply (with) / Conform (to)**
Sympathetic	$[\text{Target}] + \textbf{to} + \text{Idea}$	Empathetic (relates to emotions, not ideas)	**Receptive / Amicable**

---

3. High-Frequency Contextual Shift Map (Polysemous Anchors)

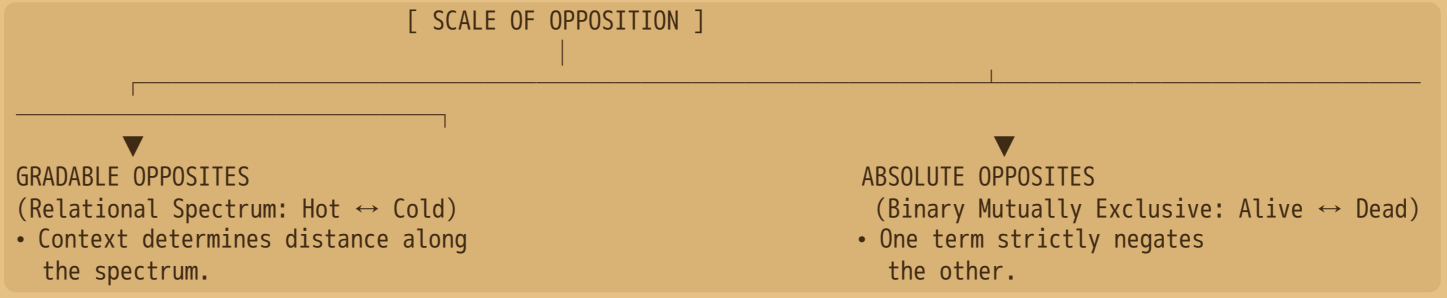
High-yield aptitude vocabulary words frequently pivot between primary everyday meanings and secondary contextual usages:

Polysemous Target	Primary / Everyday Usage	Secondary / Contextual Assessment Meaning	Contextual Synonym	Contextual Antonym
Champion	A winner of a contes t	To actively support or defend a cause/idea	**Advocate / Espouse**	Oppose / Repudia te
Compromise	Reaching a mutual a greement	To weaken, undermine, or expo se to hazard	**Impair / Endanger**	Fortify / Protect
Tender	Gentle, soft, or delic ate	To formally offer or present for acceptance	**Proffer / Submit**	Retract / Withdra w
Temper	State of anger or mo od	To soften, moderate, or dilute th e severity of	**Mitigate / Assuage**	Exacerbate / Inte nsify
Plastic	Synthetic polymer m aterial	Capable of being shaped, mold ed, or adapted	**Malleable / Impressionable**	Rigid / Inflexible
Check	To inspect or verify	To stop, slow down, or restrain progress	**Curb / Stem**	Foster / Expedite
Base	Bottom support of a n object	Devoid of moral principles; dish onorable	**Despicable / Ignoble**	Noble / Honourab le
Calculate	To solve math probl ems	Deliberate, intentional, and pre-planned	**Premeditated / Calculated**	Spontaneous / Ac cidental
Qualify	To pass a test or sta ndard	To limit, modify, or restrict a sta tement	**Cap / Restrain**	Broaden / Absolut e
Arrest	To apprehend a sus pect	To attract and hold attention; to halt growth	**Seize / Stem**	Release / Spark

---

4. Antonym Mechanics: Direct vs. Contextual Contrasts

When finding antonyms under contextual constraints, apply the **Scale of Opposition**:



1. Gradable Antonym Degree Matching Rule

If the context uses a term of moderate intensity, its contextual antonym must also be of **moderate intensity**, not an absolute extreme.

$$\text{Contextual Word: Cool} \implies \text{Correct Antonym: Warm} \quad (\text{NOT } \mathbf{Boiling})$$

$$\text{Contextual Word: Thrifty} \implies \text{Correct Antonym: Extravagant} \quad (\text{NOT } \mathbf{Profligate})$$

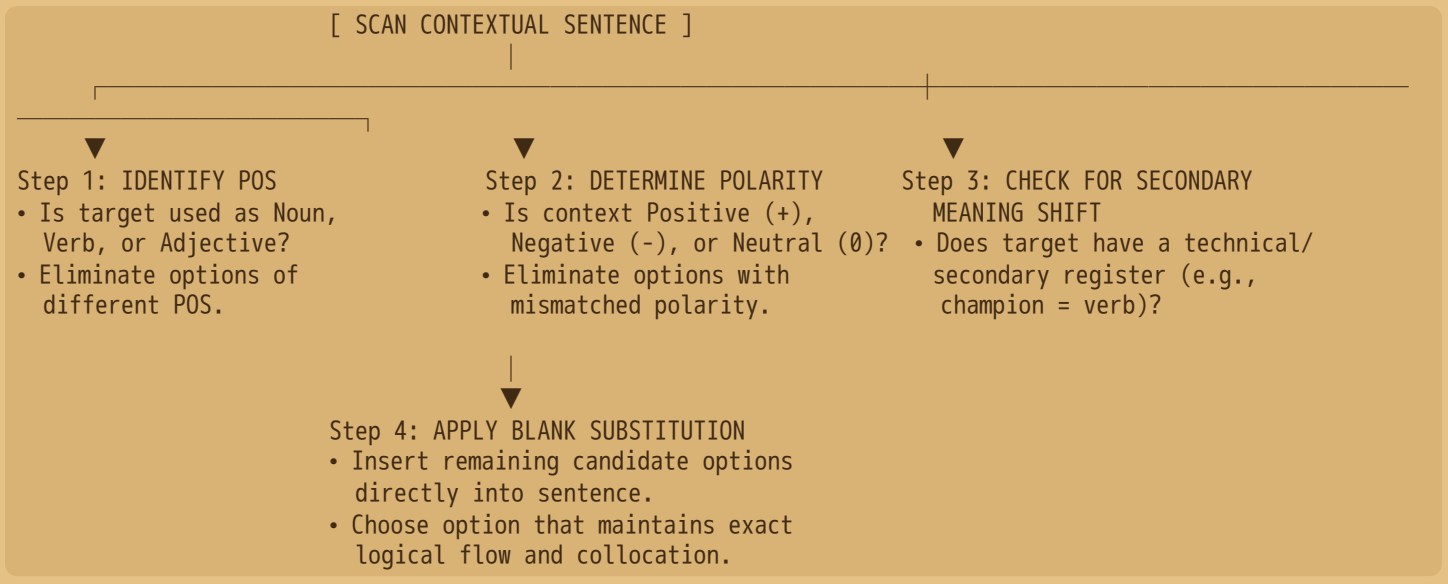
2. Contextual Shift Antonyms

In specific metaphorical contexts, a word's antonym flips completely based on the domain:

- **Context 1 (Financial Investment):** *"The firm maintains a **liquid** asset pool."*
  - $\text{Contextual Meaning} = \text{Easily convertible to cash.}$
  - $\text{Contextual Antonym} = \textbf{Illiquid/Frozen}$ (NOT "Solid" or "Firm").
- **Context 2 (Physical State):** *"The substance remained **liquid** at low temperatures."*
  - $\text{Contextual Meaning} = \text{Fluid state of matter.}$
  - $\text{Contextual Antonym} = \textbf{Solid}.$

---

5. 15-Second Contextual Elimination Protocol



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6. Common Traps & Precision Checklist

```
[ TRAP ALERT 1: Primary Meaning Bait ]
• Test setters intentionally place the PRIMARY dictionary definition as Option A.
• If the target word is simple (e.g., "Check", "Base", "Husband"), the question is
  almost certainly testing its SECONDARY or TERTIARY contextual meaning.
• NEVER pick Option A without checking if the secondary meaning fits the sentence.
```

```
[ TRAP ALERT 2: Connotation / Tone Flip ]
• Target context: "The executive's SHREWD strategy saved the division."
• "Shrewd" in this context has a POSITIVE tone (astute/perceptive).
• Option Bait: "Cunning" or "Sly" (Negative tone traps).
• Correct Match: "Astute" or "Discerning".
```

```
[ TRAP ALERT 3: Part-of-Speech Functional Shift ]
• Target context: "The board decided to SANCTION the transaction."
• "Sanction" can mean "to approve" (verb) OR "a penalty" (noun).
• Context check: Used as an infinitive verb ("to sanction") following "decided".
• Correct Synonym: "Authorize / Endorse" (NOT "Penalty" or "Embargo").
```

You

Fill in the blanks: sentence completion with two or three blank slots

D. E. Shaw Verbal Reasoning Cheat Sheet

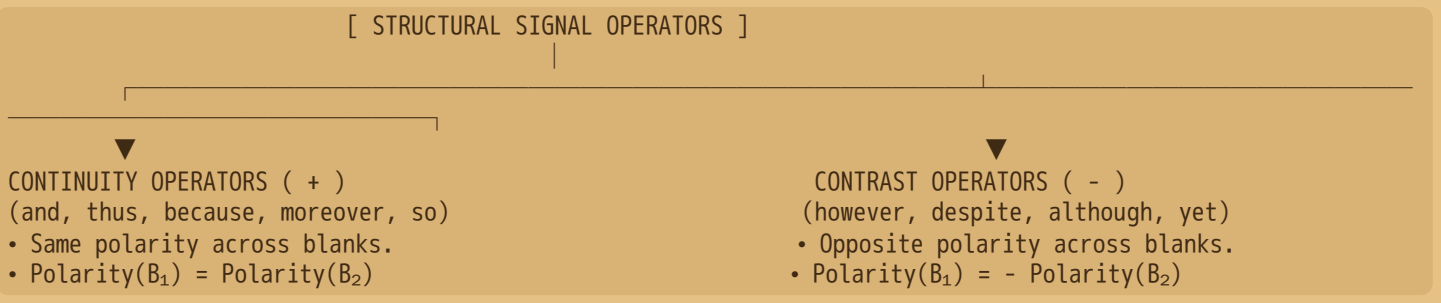
Sentence Completion: Multi-Blank Mechanics & Logical Continuity

Multi-blank sentence completion questions evaluate your ability to map sentence architecture, trace logical continuity, and align polarity across two or three interdependent slots. Success relies on **Non-Linear Solving**, **Polarity Vector Algebra**, and **Inter-Blank Relationship Linking** rather than trying options one by one.

---

1. The Polarity Vector & Inter-Blank Algebra

Sentence completion logic operates like a system of signed equations. Assign a **Polarity Vector ($+$, $-$, or 0)** to each blank based on sentence structural triggers.



Algebraic Rules for Multi-Blank Relations

$$\text{Continuity Signal: } \text{Polarity}(B_1) \times (+1) = \text{Polarity}(B_2)$$

$$\text{Contrast Signal: } \text{Polarity}(B_1) \times (-1) = \text{Polarity}(B_2)$$

$$\text{Double Negative / Negated Contrast: } (-1) \times (-1) = (+1) \quad (\text{Restores Positive Continuity})$$

- *Example:* "Far from being **[Blank 1] (-)**, the executive's report was remarkably **[Blank 2] (+)**."
  - *Trigger:* "Far from being" acts as a negative multiplier (-1).
  - If Blank 1 is negative (e.g., *flawed*), Blank 2 **must** be positive (e.g., *thorough*).

---

2. Structural Signal & Pivot Matrix

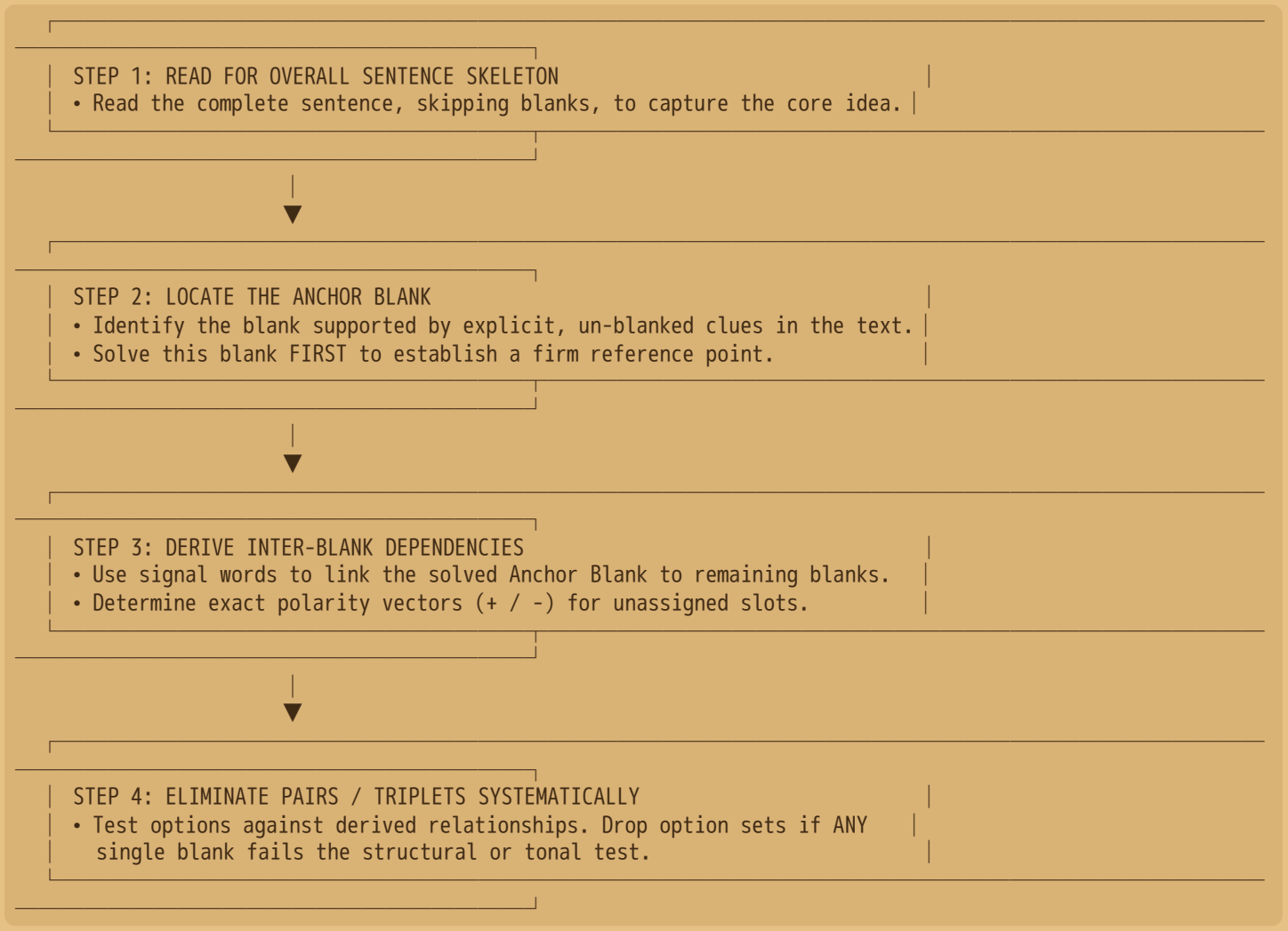
Identify structural pivot words to determine the exact relationship between missing terms before looking at the answer choices:

Signal Category	Trigger Phrases	Structural Logic	Multi-Blank Operator
Contrast / Pivot	*However, Yet, Although, Despite, Far from, On the contrary, Paradoxically*	Flips the direction or tone between clauses.	$\text{Polarity}(B_2) = -\text{Polarity}(B_1)$
Causation / Result	*Because, Thus, Consequently, Driven by, Yielding, Therefore, As a result*	Blank 2 is the direct consequence of Blank 1.	$\text{Direction}(B_1) \implies \text{Direction}(B_2)$
Elaboration / Support	*Indeed, In fact, Furthermore, Moreover, In other words, Colon (:) / Semicolon (;)*	Blank 2 defines, intensifies, or restates Blank 1.	$\text{Meaning}(B_2) \approx \text{Meaning}(B_1)$
Concession /	*Granted that, Even while, Admittedly, Notwithstanding*	Acknowledges an opposing point without changing main th	Sets up a secondary contrast shift.

Signal Category	Trigger Phrases	Structural Logic	Multi-Blank Operator
Offset		esis.	

3. The Non-Linear Multi-Pass Resolution Algorithm

Never solve a multi-blank sentence from left to right ($B_1 \rightarrow B_2 \rightarrow B_3$). Always locate the **Anchor Blank** first.



4. Multi-Blank Structural Blueprints

Blueprint 1: The Interlocking Twin-Blank (Dual Contrast)

Pattern: Clause 1 sets up an expectation; Clause 2 subverts it using a contrast pivot.

> *"Although the board initially viewed the startup's financial projections as **[Blank 1]**, the eventual quarterly returns proved the founder's strategy to be **[Blank 2]**."*

- **Clue Analysis:** "Although" signals opposite polarity between B_1 and B_2.
- **Logical Derivation:**
  - If B_1 is negative (e.g., *dubious*, *unrealistic*), B_2 must be positive (e.g., *sound*, *visionary*).
  - If B_1 is positive (e.g., *promising*), B_2 must be negative (e.g., *disastrous*).
- **Matching Pair:** $(B_1 : \text{Dubious},\ B_2 : \text{Sound})$.

Blueprint 2: The Cause-Effect Cascade (Three Blanks)

Pattern: B_1 causes B_2, which leads to an overall outcome described by B_3.

$$\text{Condition } [B_1] \xrightarrow{\text{Triggers}} \text{Action } [B_2] \xrightarrow{\text{Results In}} \text{Outcome } [B_3]$$

> *"The chief executive's **[Blank 1]** stance during negotiations **[Blank 2]** the labor union, thereby **[Blank 3]** any hope of an immediate compromise."*

- **Logical Derivation:**
  - If $B_1 = \text{Intransigent}$ (-), it will $B_2 = \text{Antagonize}$ (-) the union.
  - Antagonizing the union results in $B_3 = \text{Extinguishing} \ / \ \text{Precluding}$ (-) compromise.
- **Polarity Vector:** $(-) \rightarrow (-) \rightarrow (-)$ (Negative Cascade).

Blueprint 3: The Restatement / Colon Blueprint

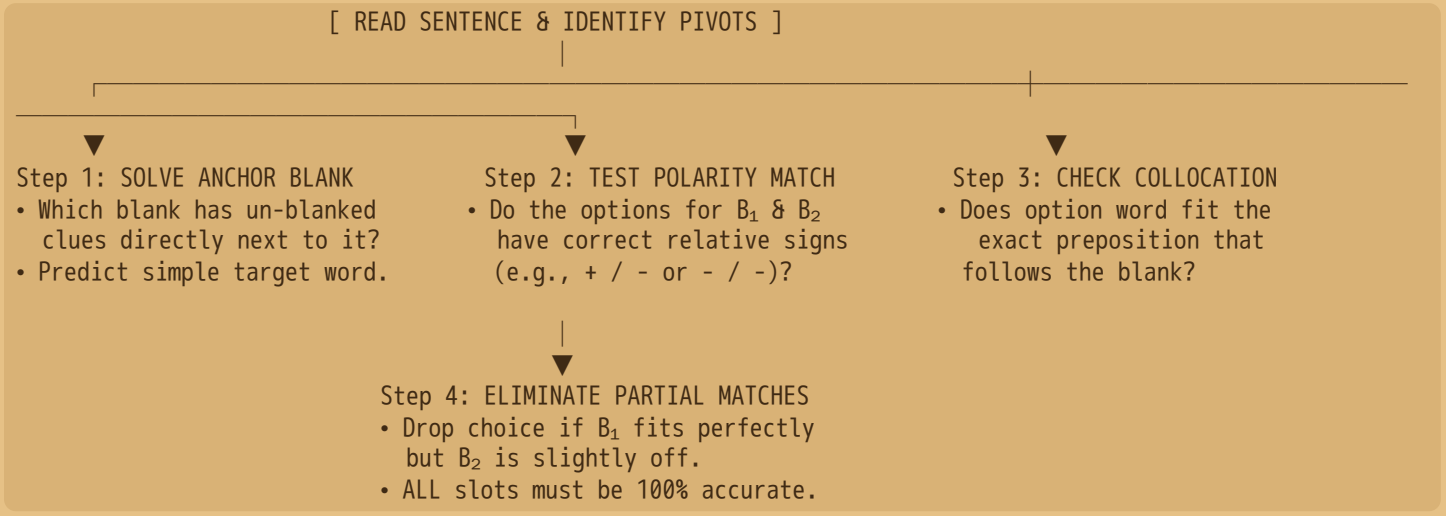
Pattern: A colon (`:`) or semicolon (`;`) indicates that the second clause explains or defines the first.

> *"The research team's findings were fundamentally **[Blank 1]**: the data not only **[Blank 2]** long-held theoretical assumptions but also **[Blank 3]** an entirely new paradigm."*

- **Clue Analysis:** The colon signals that B_2 and B_3 explain B_1.
- **Logical Derivation:**
  - B_2 ("refuted / overturned") and B_3 ("established / birthed") together describe a revolutionary outcome.
  - Therefore, B_1 must mean *groundbreaking*, *transformative*, or *paradigm-shifting*.

---

5. 15-Second Multi-Blank Elimination Protocol



---

6. Common Traps & Precision Checklist

```
[ TRAP ALERT 1: The "Fits Blank 1, Fails Blank 2" Trap ]
• Test setters design option sets where the first word is an IRRESISTIBLE fit for Blank 1,
  but the second word is logically incompatible with Blank 2.
• RULE: Never select an option based on Blank 1 alone. Always verify the LAST blank first
  or check all slots simultaneously.
```

```
[ TRAP ALERT 2: Misinterpreting Secondary Contrast Operators ]
• Phrases like "notwithstanding", "for all", and "blunted by" act as subtle contrast pivots.
  - "For all his supposed [Blank 1: expertise], his analysis was surprisingly [Blank 2: shallow]."
  - "For all X" = "Despite X". It forces a contrast between Blank 1 and Blank 2.
```

```
[ TRAP ALERT 3: Tone Shift & False Intensity ]
• Example: A blank requires a neutral word, but choices offer extreme emotional terms.
• Context: "The audit revealed **[Blank]** accounting discrepancies."
  - Incorrect: *Heinous* (Too dramatic/emotional for formal financial context).
  - Correct: *Significant* or *Material* (Matches formal analytical register).
```

- **Prepositional Collocation Check:** Always check the word immediately following the blank. If the text says `[Blank 1] to`, ensure your chosen word pairs correctly with "to" (e.g., *indifferent to*, *conducive to*) rather than "with" or "of".

You

Para jumbles: rearranging sentences into coherent paragraphs

Gemini 3.6 Thinking

D. E. Shaw Verbal Reasoning Cheat Sheet

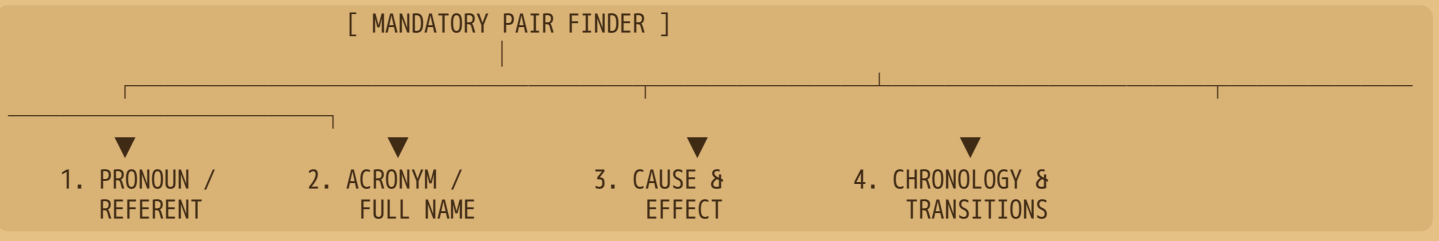
Para Jumbles: Rearranging Sentences into Coherent Paragraphs

Para jumble questions evaluate your structural logic, reading speed, and ability to track discourse markers. Cracking these under tight time constraints requires ignoring how individual sentences "sound" in isolation and using strict structural rules: **First-Principles Anchoring**, **Mandatory Pair Identification**, and **Chronological/Logical Flow Mechanics**.

---

1. Core Structural Mechanics: The 4 Structural Links

Every coherent paragraph relies on structural connective tissue. Use these four link types to bind sentences into non-negotiable pairs:



1. Pronoun & Noun Referent Chain

A pronoun (*he, she, it, they, these, this, those*) or generic noun (*the company, the process*) cannot appear without its antecedent being introduced earlier.

$$\text{Full Noun / Entity Introduction } (A) \longrightarrow \text{Pronoun / Referent } (B) \implies [\mathbf{A} - \mathbf{B}]$$

- *Example:*
  - (Sentence X): *"**These financial disruptions** forced central banks to act."*
  - (Sentence Y): *"In 2008, **unprecedented liquidity crises** hit international markets."*
  - *Link:* Sentence Y introduces the specific crisis, while Sentence X references it via "These financial disruptions" $\implies [\mathbf{Y} - \mathbf{X}]$.

2. Name / Acronym Expansion Protocol

Introductions precede abbreviations, shortened names, or acronyms:

$$\text{Full Name / Title} \longrightarrow \text{Surname / Acronym}$$

- **Full Title:** *"The World Health Organization (WHO)..."* $\longrightarrow$ **Acronym:** *"The WHO..."*
- **Full Name:** *"Dr. Radhika Gupta..."* $\longrightarrow$ **Surname:** *"Gupta further stated..."*

---

3. Transition & Discourse Marker Mechanics

Transition words reveal the exact structural role of a sentence:

Transition Type	Trigger Words	Structural Position & Function
Introductory / Scope	*Historically, Traditionally, In recent years, For decades*	Sets background; prime candidate for **Opening Sentence (S_1)**.
Contrast / Pivot	*However, Nevertheless, On the other hand, Yet, Conversely*	Pivots away from previous idea. **Cannot start a paragraph** unless establishing a universal counter-premise.
Elaboration / Addition	*Furthermore, Moreover, In addition, Similarly, Indeed*	Continues an existing thought. Must immediately follow a related premise.
Conclusion / Result	*Thus, Therefore, Consequently, Ultimately, As a result*	Summarizes or states the final consequence. Prime candidate for **Closing Sentence (S_n)**.

---

4. General-to-Specific Hierarchy (Deductive Flow)

Standard non-fiction paragraphs move from **broad, overarching concepts** to **specific applications, instances, or examples**:

$$\text{Broad Concept / Definition} \longrightarrow \text{Specific Mechanism} \longrightarrow \text{Concrete Example / Case}$$

---

2. Opening & Closing Isolation Protocols

Isolation Protocol: The First Sentence (S_1)

The opening sentence of a paragraph is **standalone and self-contained**.

```
┌──────────────────────────────────────────────────────────────────────┐
│ OPENING SENTENCE ($S_1$) CHECKLIST                                   │
│ • Does NOT start with explicit contrast transitions (However, Yet, Thus). │
│ • Does NOT contain unreferenced pronouns (They, These, This process).      │
│ • Introduces the core subject/topic without assuming prior knowledge.     │
│ • Sets the broad context or defines the central domain.                   │
└──────────────────────────────────────────────────────────────────────┘
```

Isolation Protocol: The Closing Sentence (S_n)

The final sentence synthesizes the argument, presents a final outcome, or projects future implications.

```
┌──────────────────────────────────────────────────────────┐
│ CLOSING SENTENCE ($S_n$) CHECKLIST                       │
│ • Contains conclusive transition markers (Thus, Therefore, Ultimately).  │
│ • Summarizes the broad consequence of the discussed topic.              │
│ • Does NOT introduce a completely new, unexplained concept.             │
└──────────────────────────────────────────────────────────┘
```

---

3. The 4-Step Resolution Algorithm

```
┌──────────────────────────────────────────────────────────┐
│ STEP 1: SCAN FOR INDEPENDENT ANCHOR ($S_1$)              │
│ • Read all options quickly. Eliminate sentences with pronouns/transitions │
│   to locate the unique opening sentence ($S_1$).         │
└──────────────────────────────────────────────────────────┘
                          │
                          ▼
┌──────────────────────────────────────────────────────────┐
│ STEP 2: BUILD MANDATORY PAIRS                            │
│ • Look for Noun-Pronoun chains, Name-Acronym pairs, or Cause-Effect links. │
│   (e.g., Sentence B MUST follow Sentence D ──► Block [DB]).  │
└──────────────────────────────────────────────────────────┘
                          │
                          ▼
┌──────────────────────────────────────────────────────────┐
│ STEP 3: MAP TRANSITION DIRECTIONS                        │
│ • Check contrast markers (However) to see which two blocks are being  │
│   compared. Position positive/negative sides accordingly.  │
└──────────────────────────────────────────────────────────┘
                          │
                          ▼
┌──────────────────────────────────────────────────────────┐
│ STEP 4: VERIFY AGAINST ANSWER CHOICES / SEQUENTIAL LOGIC │
│ • Match your Mandatory Blocks (e.g., [DB], [AC]) against the options.  │
│ • Read the reconstructed sequence end-to-end to confirm logical flow.  │
└──────────────────────────────────────────────────────────┘
```

---

4. Worked Structural Pattern Blueprints

Pattern 1: Problem-Solution Sequence

- **Structure:** Identify Problem $\longrightarrow$ Explain Impact $\longrightarrow$ Propose Solution $\longrightarrow$ Evaluate Outcome
- **Example Chain:**
  1. *(S1)* Cyberattacks on financial infrastructure have reached an all-time high.
  2. *(S2)* These breaches cost institutions billions and erode public trust.
  3. *(S3)* To mitigate these threats, banks are deploying zero-trust security architectures.
  4. *(S4)* As a result, unauthorized access attempts have dropped significantly.

Pattern 2: Chronological / Historical Progression

- **Structure:** $\text{Past State} \longrightarrow \text{Trigger / Shift} \longrightarrow \text{Present State} \longrightarrow \text{Future Outlook}$
- **Key Markers:** Dates (*In 1990, By 2015*), time phrases (*Initially, Subsequently, Currently*).

Pattern 3: Hypothesis-Evaluation Sequence

- **Structure:** $\text{Introduce Theory/Claim} \longrightarrow \text{Provide Supporting Data} \longrightarrow$ $\text{Introduce Counter-Evidence} \longrightarrow \text{Synthesize Conclusion}$

---

5. 15-Second Elimination Matrix (Using Mandatory Pairs)

When options are provided (e.g., A) BACD, B) ACDB, C) DBAC, D) BADC):

1. Identify **ONE definite mandatory pair** (e.g., You know for certain that sentence **D** must be followed by **B** $\rightarrow [\mathbf{DB}]$).
2. Scan options for the substring `DB`:
   - A) B A C D $\rightarrow$ (No `DB`) $\rightarrow$ **Eliminate**
   - B) A C D B $\rightarrow$ (Contains `DB`) $\rightarrow$ **Keep**
   - C) D B A C $\rightarrow$ (Contains `DB`) $\rightarrow$ **Keep**
   - D) B A D C $\rightarrow$ (No `DB`) $\rightarrow$ **Eliminate**
3. Locate the opening sentence (S_1) between options B (Starts with **A**) and C (Starts with **D**).
4. If **D** contains a pronoun referencing **A**, **D** cannot be first $\rightarrow$ **Correct Choice: B**.

---

6. Common Traps & Precision Checklist

```
[ TRAP ALERT 1: The "Sounds Okay" Narrative Fallacy ]
• Do not rely on reading the paragraph aloud in your head to check if it "sounds smooth".
• Test setters construct wrong options that sound fluent but violate structural logic
  (e.g., placing a pronoun before its antecedent).
```

```
[ TRAP ALERT 2: Misinterpreting "This" or "Such" as Starters ]
• Sentences starting with "This phenomenon", "Such strategies", or "These results"
  are NEVER opening sentences.
• They require an immediately preceding sentence that explicitly defines the
  phenomenon, strategy, or result.
```

```
[ TRAP ALERT 3: Chronological Trap (Flashbacks) ]
• A passage usually moves forward in time (1980 ——► 2000 ——► 2020).
• TRAP: A sentence starting with "Retrospectively" or "Looking back..." can jump
  backward in time, but it must logically follow the present observation.
```

- **Definite Article Trap:** *"A market crash occurred..."* (Indefinite article `A` = **Introduction / First mention**). *"The market crash caused..."* (Definite article `The` = **Subsequent mention**). Always place `A` before `The`.